Polarization of Gravitational Waves From Turbulence: Tension Between Simulation and Analytic Approaches

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I. GRAVITATIONAL WAVES FROM A TURBULENT SOURCE

Before specializing to particular spectral and temporal models, we collect the standard analytic derivation of the gravitational-wave (GW) spectrum sourced by turbulence [1–5]. This fixes the conventions used throughout and, in particular, shows why the remaining sections reduce the problem to computing the contracted source correlator $H_{ijij}(\mathbf{k},\omega)$ of Eq. (11) on the diagonal $\omega = k$.

A. Sourced wave equation

Write the tensor perturbations of a spatially flat FLRW background as

$$ds^2 = a^2(\tau)\big[-d\tau^2 + (\delta_{ij} + h_{ij})\, dx^i dx^j\big], \qquad \partial_i h_{ij} = h_{ii} = 0, \tag{1}$$

with h_{ij} transverse-traceless (TT). The linearized Einstein equations give

$$h''_{ij} + 2\frac{a'}{a}\, h'_{ij} - \nabla^2 h_{ij} = 16\pi G\, a^2\, \Pi^{\mathrm{TT}}_{ij}, \tag{2}$$

where $' = d/d\tau$ and Π^{TT}_{ij} is the TT part of the anisotropic stress. For an incompressible, non-relativistic turbulent fluid the source is the Reynolds stress,

$$\Pi^{\mathrm{TT}}_{ij}(\mathbf{k},\tau) = \Lambda_{ij,lm}(\hat{k})\, T_{lm}(\mathbf{k},\tau), \qquad T_{lm} = w\, v_l v_m, \qquad \Lambda_{ij,lm} = P_{il}P_{jm} - \tfrac{1}{2}P_{ij}P_{lm}, \tag{3}$$

with $w = \bar{\rho} + \bar{p}$ the enthalpy density, $P_{ij} = \delta_{ij} - \hat{k}_i\hat{k}_j$, and Λ the TT projector. The double TT projection together with the Gaussian (Wick) reduction of the quartic velocity correlator collapses the two-point function of (3) onto the contracted form $R_{ijij} = R_{ii}R_{jj} + \frac{1}{3}R_{ij}R_{ij}$ that appears in Eq. (11).

Turbulence acts on eddy-turnover times short compared with a Hubble time, so over the source lifetime we drop the expansion ($a \approx 1$, friction negligible) and Eq. (2) becomes, in Fourier space, a forced oscillator,

$$h''_{ij}(\mathbf{k},\tau) + k^2 h_{ij}(\mathbf{k},\tau) = 16\pi G\, \Pi^{\mathrm{TT}}_{ij}(\mathbf{k},\tau). \tag{4}$$

B. Green-function solution and the GW spectrum

The retarded solution of Eq. (4) and its time derivative are

$$h_{ij}(\mathbf{k},\tau) = \frac{16\pi G}{k}\int_{\tau_i}^{\tau} d\tau'\, \sin[k(\tau-\tau')]\, \Pi^{\mathrm{TT}}_{ij}(\mathbf{k},\tau'), \qquad \dot{h}_{ij}(\mathbf{k},\tau) = 16\pi G\int_{\tau_i}^{\tau} d\tau'\, \cos[k(\tau-\tau')]\, \Pi^{\mathrm{TT}}_{ij}(\mathbf{k},\tau'). \tag{5}$$

With the Isaacson energy density $\rho_{\mathrm{GW}} = \langle \dot{h}_{ij}\dot{h}_{ij}\rangle/(32\pi G)$ and the unequal-time correlator (UETC) of the TT stress,

$$\langle \Pi^{\mathrm{TT}}_{ij}(\mathbf{k},\tau_1)\, \Pi^{\mathrm{TT}*}_{ij}(\mathbf{k}',\tau_2)\rangle = (2\pi)^3\, \delta^{(3)}(\mathbf{k}-\mathbf{k}')\, \Pi(k;\tau_1,\tau_2), \tag{6}$$

inserting (5) and averaging the product $\cos[k(\tau-\tau_1)]\cos[k(\tau-\tau_2)] \to \frac{1}{2}\cos[k(\tau_1-\tau_2)]$ over the fast GW oscillations gives the master formula

$$\frac{d\rho_{\rm GW}}{d\ln k} \simeq \frac{2G\,k^3}{\pi}\int d\tau_1 \int d\tau_2\, \cos[k(\tau_1-\tau_2)]\,\Pi(k;\tau_1,\tau_2). \tag{7}$$

The GW spectrum is thus the temporal Fourier transform of the source UETC, evaluated at the on-shell frequency $\omega = k$. For a statistically stationary (or quasi-stationary) source, $\Pi(k;\tau_1,\tau_2) = \Pi(k;\tau_1-\tau_2)$ and the double time integral factorizes into the source duration $T_{\rm src}$ times the temporal transform $\widetilde{\Pi}(k,\omega) = \int d(\Delta\tau)\, e^{i\omega\Delta\tau}\Pi(k;\Delta\tau)$, which is precisely the spatio-temporal correlator $H_{ijij}(\mathbf{k},\omega)$ of Eq. (11). Hence

$$\frac{d\rho_{\rm GW}}{d\ln k} \;\propto\; k^3\, H_{ijij}(\mathbf{k},\omega)\big|_{\omega=k} \quad\Longrightarrow\quad \Omega_{\rm GW}(p) \;\propto\; p^3\, H(p,p), \qquad p = \frac{k}{k_0},\quad q = \frac{\omega}{k_0}, \tag{8}$$

the GW dispersion $\omega = k$ becoming the diagonal $q = p$ used in Sec. X (with characteristic strain $h_c^2(p) = p\,H(p,p)$). Everything downstream is therefore an evaluation of H_{ijij}, whose spatial part is a convolution of the velocity power spectrum (quadratic source, Eqs. (12)–(13)) and whose temporal part is the decorrelation model carried by $\Pi(k;\tau_1,\tau_2)$.

C. Spectral shape from the literature

For Kolmogorov inertial-range input the analytic results have three robust pieces:

- *Infrared ($k \ll k_0$):* causality forces the stress correlator to be analytic (white) as $k \to 0$, so $H \to$ const and $\Omega_{\rm GW} \propto k^3$ [3, 4].
- *Peak ($k \sim k_0$):* set by the largest eddies, with parametric amplitude $(d\Omega_{\rm GW}/d\ln k)_{\rm peak} \sim (H_*/k_0)\,\Omega_K^2$, i.e. the squared kinetic-energy fraction suppressed by the source-duration factor $H_*/k_0 \sim H_*\tau_{\rm eddy}$ [1, 3].
- *Ultraviolet ($k \gg k_0$): model-dependent*, fixed by the temporal decorrelation $\Pi(k;\tau_1,\tau_2)$. Kraichnan sweeping gives the Gogoberidze tail [2], while a δ-correlated or a decaying-turbulence kernel give different slopes.

The UV slope is precisely where the four UETC models derived below (δ-spectrum vs. white-noise, Kraichnan vs. decay) differ, which is the focus of the remainder of the paper.

D. The model space, and a uniform notation for the kernels

Every kernel derived in this paper is a choice of two independent ingredients, and it is worth making the factorisation explicit at the outset because most of the disagreements in the literature—and every discrepancy discussed below—trace to one axis or the other rather than to the derivation itself.

a. Axis 1: the spatial source model S. What the fluid two-point function is taken to be. We use four:

- $S_{\rm K41}$: Kolmogorov inertial range, $E(k) \propto k^{-5/3}$ on $k_0 < k < k_d$, with $R = (k_d/k_0)^{4/3}$ the Reynolds number (Sec. II);
- $S_{\delta k}$: monochromatic, $E(k) = E_0\delta(k-k_0)$ (Sec. V A);
- $S_{\delta r}$: real-space white noise, $R_{ij} \propto \delta^{(3)}(\mathbf{r})$, equivalently $E(k) \propto k^2$ (Sec. V B);
- $S_{\rm full}$: a full input spectrum—a Batchelor or Saffman infrared band joined to the Kolmogorov inertial range (Sec. II D).

b. Axis 2: the temporal model T. What the unequal-time correlator (UETC) of the source is taken to be. We use four:

- $T_{\rm sw}$: Kraichnan sweeping, a *Gaussian* in the time difference, $f(\tau) = \exp(-\frac{\pi}{4}\eta_k^2\tau^2)$ with $\eta_k \propto \varepsilon^{1/3}k^{2/3}$;
- $T_{\rm dec}$: BK2016 decaying turbulence, a *power law* $f(\tau) = (1+|\tau|/\tau_k)^{-2/3}$;
- $T_{\rm imp}$: impulsive, δ-correlated in time (Sec. VI);
- $T_{\rm burst}$: a source of finite duration Δt (Sec. VII B).

c. Notation. We write the resulting dimensionless kernel as

$$H[S,T](p,q;M,R), \qquad p=\frac{k}{k_0}, \quad q=\frac{\omega}{k_0}, \tag{9}$$

suppressing M or R where the kernel does not depend on them, and the GW spectrum on the sound-cone diagonal is always $\Omega_{\rm GW}(p) \propto p^3 H[S,T](p,p)$. Table I lists every combination derived here, its defining equation, and its analytic structure; Sec. IX then compares them systematically. Two entries of the grid deserve a warning before any spectrum is read off them: $S_{\delta k}$ has zero spectral width and $S_{\delta r}$ has no infrared power at all, so both violate the hypotheses of the universal infrared theorem (Sec. IV A) and neither should be expected to show k^3.

Kernel	Defining equation	Analytic structure	Sec.
$H[S_{\rm K41}, T_{\rm sw}]$	Eq. (22)	2-D wavevector integral, Gaussian temporal factor	II
$H[S_{\rm K41}, T_{\rm dec}]$	Eq. (33)	2-D wavevector × 1-D Feynman parameter	III B
$H[S_{\delta k}, T_{\rm sw}]$	Eq. (46)	closed form; M in the frequency scale only	V A 1
$H[S_{\delta k}, T_{\rm dec}]$	Eq. (47)	closed form, Eq. (49)	V A 2
$H[S_{\delta r}, T_{\rm sw}]$	Eq. (54)	2-D wavevector integral; M-cutoff in the bounds	V B
$H[S_{\delta r}, T_{\rm dec}]$	Eq. (55)	2-D wavevector × Eq. (57)	V B
$H[S_{\delta k}, T_{\rm imp}]$	Eq. (70)	closed form; no M or q dependence	VI
$H[S_{\delta k}, T_{\rm burst}]$	Eq. (86)	closed form; window factor $\sin^2$	VII B
$H[S_{\rm full}, T_{\rm sw}]$	Eq. (24)	as $S_{\rm K41}$ with a shape factor $S(k)$	II D

TABLE I. The model grid. Each kernel is a (spatial source, temporal UETC) pair in the notation of Eq. (9). The analytic structure column is what actually has to be evaluated: note that the two decaying entries are closed-form or single-integral rather than the frequency convolutions they are usually written as, by Eqs. (49) and (57).

II. KRAICHNAN–K41 MODEL (GOGOBERIDZE 2007)

We use the spatial/temporal Fourier pair $R_{ij}(\mathbf{r},\tau) = \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{r}} F_{ij}(\mathbf{k},\tau)$ and $f(\eta_k,\tau) = \int \frac{d\omega}{2\pi} e^{-i\omega\tau} \tilde{f}(\eta_k,\omega)$, and adopt the isotropic incompressible model of [2] (Eqs. 23–26),

$$F_{ij}(\mathbf{k},\tau) = A(k)\, P_{ij}(\hat{k})\, f(\eta_k,\tau), \qquad P_{ij}(\hat{k}) = \delta_{ij} - \hat{k}_i\hat{k}_j, \tag{10}$$

with $A(k) = C_k \epsilon^{2/3} k^{-11/3}/4\pi$, $f(\eta_k,\tau) = \exp(-\frac{\pi}{4}\eta_k^2\tau^2)$, and $\eta_k = \frac{1}{\sqrt{2\pi}}\epsilon^{1/3}k^{2/3}$.

The fourth-rank Fourier-space correlator $H_{ijkl}(\mathbf{k},\omega) = (2\pi)^{-4}\int d\tau\, d^3\boldsymbol{\xi}\, e^{i\omega\tau - i\mathbf{k}\cdot\boldsymbol{\xi}} R_{ijkl}(\boldsymbol{\xi},\tau)$ with the contraction $R_{ijij} = R_{ii}R_{jj} + \frac{1}{3}R_{ij}R_{ij}$ (Gogoberidze Eq. 28) reads, in real space,

$$H_{ijij}(\mathbf{k},\omega) = \frac{1}{(2\pi)^4}\int d\tau\, d^3\mathbf{r}\, e^{i\omega\tau - i\mathbf{k}\cdot\mathbf{r}} \Big[R_{ii}(\mathbf{r},\tau)R_{jj}(\mathbf{r},\tau) + \tfrac{1}{3}R_{ij}(\mathbf{r},\tau)R_{ij}(\mathbf{r},\tau)\Big]. \tag{11}$$

Inserting the spatial Fourier representation of R_{ij}, the $\mathbf{r}$ integral produces a momentum-conserving $\delta^{(3)}(\mathbf{k}-\mathbf{q}-\mathbf{p})$. Renaming $\mathbf{q} \to \mathbf{k}_1$, $\mathbf{u} = \mathbf{k} - \mathbf{k}_1$,

$$H_{ijij}(\mathbf{k},\omega) = \frac{1}{(2\pi)^7}\int d\tau\, e^{i\omega\tau}\, d^3k_1 \left[F_{ii}(\mathbf{k}_1,\tau)F_{jj}(\mathbf{u},\tau) + \tfrac{1}{3}F_{ij}(\mathbf{k}_1,\tau)F_{ij}(\mathbf{u},\tau)\right]. \tag{12}$$

The temporal Fourier transform then converts the τ integral into a frequency δ-function, leaving the frequency convolution

$$H_{ijij}(\mathbf{k},\omega) = \frac{1}{(2\pi)^8}\int d^3k_1\, d\omega_1 \left[\tilde{F}_{ii}(\mathbf{k}_1,\omega_1)\tilde{F}_{jj}(\mathbf{u},\omega-\omega_1) + \tfrac{1}{3}\tilde{F}_{ij}(\mathbf{k}_1,\omega_1)\tilde{F}_{ij}(\mathbf{u},\omega-\omega_1)\right]. \tag{13}$$

Substituting Eq. (10) and using $P_{ii}(\hat{k}) = 2$ together with $P_{ij}(\hat{k}_1)P_{ij}(\hat{u}) = 1 + (\hat{k}_1\cdot\hat{u})^2$, the tensor contractions reduce Eq. (13) to

$$H_{ijij}(\mathbf{k},\omega) = \frac{1}{(2\pi)^8}\int d^3k_1\, d\omega_1\, A(k_1)A(u)\, \tilde{f}(\eta_{k_1},\omega_1)\tilde{f}(\eta_u,\omega-\omega_1) \left[4 + \tfrac{1}{3}\big(1 + (\hat{k}_1\cdot\hat{u})^2\big)\right]. \tag{14}$$

This is the Fourier representation valid for all k; we treat $\mathbf{k} > \mathbf{k}_0$ and $\mathbf{k} \le \mathbf{k}_0$ separately below.

A. $\mathbf{k} > \mathbf{k}_0$

The law of cosines $\hat{k}_1 \cdot \hat{u} = (k^2 - k_1^2 - u^2)/(2k_1 u)$ turns the bracket in Eq. (14) into a homogeneous polynomial in (k, k_1, u); expanding and grouping gives the geometric kernel

$$\mathcal{K}(k, k_1, u) \equiv \frac{27}{6} + \frac{k^4}{12k_1^2u^2} + \frac{k_1^2}{12u^2} + \frac{u^2}{12k_1^2} - \frac{k^2}{6u^2} - \frac{k^2}{6k_1^2} \qquad \text{(cf. Eq. 29 of [2]).} \tag{15}$$

The temporal transform $\tilde{f}(\eta_k, \omega) = (2/\eta_k)\exp(-\omega^2/\pi\eta_k^2)$ combines with $A(k)$ into the single spectral factor

$$g(k, \omega) \equiv A(k)\,\tilde{f}(\eta_k, \omega) = \frac{2E_k}{4\pi k^2 \eta_k}\exp\left(-\frac{\omega^2}{\pi\eta_k^2}\right), \qquad E_k = C_k \epsilon^{2/3} k^{-5/3}. \tag{16}$$

Going to spherical coordinates for $\mathbf{k}_1$ ($d^3k_1 = k_1^2\,dk_1\,d\phi\,d\mu$ with $\mu = \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}_1$), the azimuthal integral contributes 2π. Trading μ for $u = \sqrt{k^2 + k_1^2 - 2kk_1\mu}$ via $d\mu = u\,du/(kk_1)$, with $u \in [|k - k_1|, k + k_1]$,

$$H_{ijij}(\mathbf{k}, \omega) = \frac{1}{(2\pi)^7}\int_0^\infty \frac{k_1\,dk_1}{k}\int_{|k-k_1|}^{k+k_1} u\,du \int_{-\infty}^{\infty} d\omega_1\, g(k_1, \omega_1)\, g(u, \omega - \omega_1)\, \mathcal{K}(k, k_1, u). \tag{17}$$

Switching to the dimensionless variables $q = k_1/k$, $s = u/k$, $\Omega = \omega/(\epsilon^{1/3}k^{2/3})$, $\Omega_1 = \omega_1/(\epsilon^{1/3}k^{2/3})$, g becomes $g(k, \omega) = (1/\sqrt{2\pi})\, C_k \epsilon^{1/3} k^{-13/3}\exp(-2\Omega^2)$. The product $g(k_1, \omega_1) g(u, \omega - \omega_1)$ thus separates into a dimensional prefactor $\propto k^{-26/3}$ times a manifestly dimensionless Gaussian envelope $\Phi(q, \Omega_1; s, \Omega - \Omega_1) = q^{-13/3}s^{-13/3}\exp[-2\Omega_1^2 q^{-4/3} - 2(\Omega - \Omega_1)^2 s^{-4/3}]$. Collecting all k-dependent factors gives the scaling form

$$H_{ijij}(\mathbf{k}, \omega) = \frac{C_k^2\,\epsilon}{256\,\pi^8}\,k^{-5}\,\mathfrak{H}(\Omega), \qquad \mathfrak{H}(\Omega) = \int_0^\infty dq\, q\int_{-\infty}^{\infty} d\Omega_1 \int_{|1-q|}^{1+q} ds\, s\;\Phi\,\widetilde{\mathcal{K}}(q, s), \tag{18}$$

with $\widetilde{\mathcal{K}}(q, s) = \mathcal{K}(k, k_1, u)|_{k_1 = qk, u = sk}$ (homogeneous of degree zero). All physical scales are absorbed in the prefactor $C_k^2 \epsilon k^{-5}$.

B. Gogoberidze (2007) Appendix A form

Using the Gaussian-erfc identity (Eq. A.1 of [2], $\int_0^\infty dy\, e^{-Ay^2 - B(y-x)^2} = \sqrt{\pi/[4(A+B)]}\, e^{-ABx^2/(A+B)}\, \mathrm{erfc}[-Bx/\sqrt{A+B}]$), the ω_1 integration in Eq. (17) can be done in closed form, yielding the doubly-bounded form

$$\begin{aligned} H_{ijij}(k, \omega) = \frac{\varepsilon}{24\pi(2\pi)^{3/2}k}&\int_{k_0}^{k_d} dk_1\, k_1^{-10/3}\int_{\max(|k_1-k|, k_0)}^{\min(k_1+k, k_d)} du\, u^{-10/3}\,(k_1^{-4/3} + u^{-4/3})^{-1/2} \\ &\times\left[27 - \tfrac{k^2}{k_1^2} - \tfrac{k^2}{u^2} + \tfrac{k^4}{2k_1^2u^2} + \tfrac{k_1^2}{2u^2} + \tfrac{u^2}{2k_1^2}\right]\exp\!\left(-\tfrac{2\varepsilon^{-2/3}\omega^2}{k_1^{4/3} + u^{4/3}}\right)\mathrm{erfc}\!\left(-\tfrac{\sqrt{2}\,\varepsilon^{-1/3}\omega}{(k_1^{-4/3} + u^{-4/3})^{1/2}}\right). \end{aligned} \tag{19}$$

k_0 is the outer scale and k_d the dissipation scale; the integration window collapses near the boundaries of the inertial range. This integral is finite as $k \to 0$.

Define the dimensionless parameters

$$M = \left(\frac{\varepsilon}{k_0}\right)^{1/3}, \quad \frac{k_d}{k_0} = R^{3/4}, \quad p = \frac{k}{k_0}, \quad q = \frac{\omega}{k_0}, \qquad x = \left(\tfrac{k_1}{k_0}\right)^{-4/3}, \quad y = \left(\tfrac{u}{k_0}\right)^{-4/3}, \tag{20}$$

where $M = (\varepsilon/k_0)^{1/3}$ is the outer-scale turbulent Mach number (sound speed set to unity), equal to the rms eddy velocity u_0. The Mach number enters the GW kernel through the k-dependent sweeping rate $\eta_k = \frac{1}{\sqrt{2\pi}}\varepsilon^{1/3}k^{2/3} = \frac{M}{\sqrt{2\pi}}k_0^{1/3}k^{2/3}$, which couples frequency to wavenumber in the Gaussian factor of Eq. (19). As a result M sets the *effective* wavenumber support of the integral: the exponential $\exp[-2q^2/(M^2(z^{4/3} + y^{4/3}))]$ (with $z = k_1/k_0$) cuts off contributions with $z^{4/3} + y^{4/3} \lesssim 2q^2/M^2$, so the peak of the GW spectrum shifts with M. The substitution $k_1 = k_0 x^{-3/4}$, $u = k_0 y^{-3/4}$ gives $dk_1\, k_1^{-10/3} = -(3/4)k_0^{-7/3}x^{3/4}dx$ and $(k_1^{-4/3} + u^{-4/3})^{-1/2} = k_0^{2/3}(x + y)^{-1/2}$; the integration limits become $x \in [R^{-1}, 1]$ and $y \in [y_{\min}(x), y_{\max}(x)]$ with

$$y_{\max}(x) = \min(|x^{-3/4} - p|^{-4/3},\, 1), \qquad y_{\min}(x) = \max((x^{-3/4} + p)^{-4/3},\, R^{-1}). \tag{21}$$

Plugging into Eq. (19) yields

$$H_{ijij}(p,q) = \frac{3M^3k_0^{-4}}{256(2\pi)^{3/2}\,p}\int_{R^{-1}}^{1} dx\, x^{3/4}\int_{y_{\min}}^{y_{\max}} dy\, y^{3/4}\,(x+y)^{-1/2}$$
$$\times\left[27 - p^2(x^{3/2}+y^{3/2}) + \tfrac{p^4}{2}x^{3/2}y^{3/2} + \tfrac{1}{2}(x^{-3/2}y^{3/2} + y^{-3/2}x^{3/2})\right]$$
$$\times \exp\!\left(-\tfrac{2xy}{x+y}\tfrac{q^2}{M^2}\right)\,\mathrm{erfc}\!\left(-\tfrac{\sqrt{2}\,q}{M\sqrt{x+y}}\right). \tag{22}$$

C. Mach-dependent peak of the stationary spectrum

Evaluating the kernel (22) on the GW dispersion $\omega = k$ (the diagonal $q = p$ of Eq. (8)) gives the dimensionless energy-density spectrum $\Omega_{\rm GW}(p) \propto p^3\, H_{ijij}(p,p;M,R)$ of stationary Kraichnan–K41 turbulence. The Mach number enters *only* through the sweeping rate $\eta_k \propto M\,k^{2/3}$, i.e. through the Gaussian $\exp[-\frac{2xy}{x+y}\,q^2/M^2]$ and the accompanying erfc in Eq. (22). This factor is a soft cutoff that suppresses the integrand once $p \gtrsim M$, so the spectral peak tracks the Mach number, $p_{\rm peak} \sim M$, until it saturates near the outer scale at large M. Figure 1 shows this directly: the peak slides from $p_{\rm peak} \simeq 0.05$ at $M = 0.03$ to $p_{\rm peak} \simeq 2.6$ at $M = 3$, while the amplitude climbs steeply (the M^3 prefactor of Eq. (22) times the widening M-support of the integral). The infrared rise is the universal causal slope $\Omega_{\rm GW} \propto p^3$ (Sec. I C), common to all M; only the location and height of the peak respond to M.

The M-dependence of the IR amplitude is fixed entirely by the prefactor: for $p \ll M$ the Gaussian and erfc factors of Eq. (22) reduce to unity, so $H_{ijij}(p \to 0)$ is set by the M^3 prefactor alone and $\Omega_{\rm GW} \to C\,M^3p^3$ with C a universal (M-independent) constant. Equivalently the characteristic strain scales as $h_c = \sqrt{p\,H} \sim M^{3/2}\sqrt{p}$ in the IR. Compensating the spectrum by M^{-3} (the strain by $M^{-3/2}$) therefore collapses every curve onto the single universal p^3 line in the IR, isolating the genuinely M-dependent peak and UV; Fig. 1(b) confirms this collapse numerically (residual spread $< 1.5\%$ at $p \simeq 3\times10^{-4}$, and a measured IR exponent $d\ln H/d\ln M \to 3.00$ as $p \to 0$).

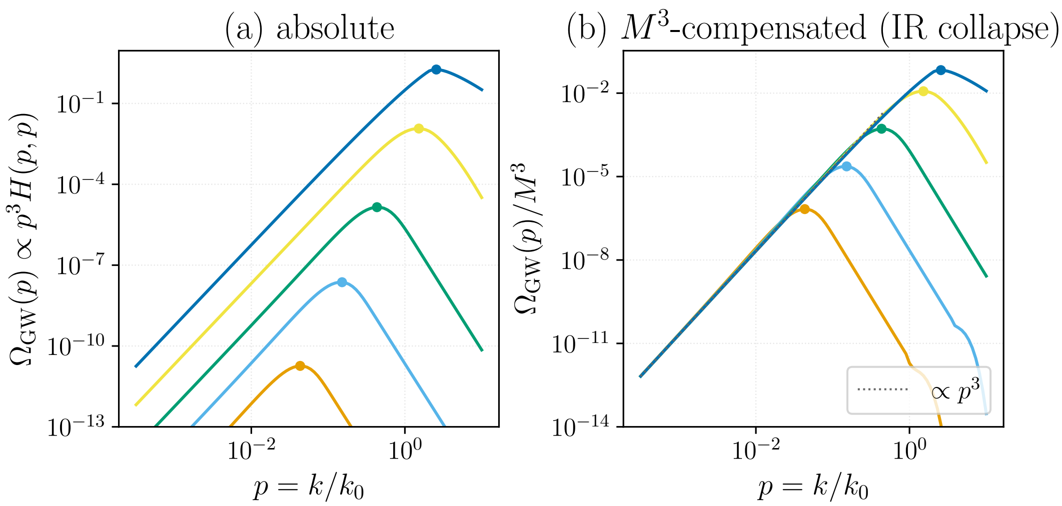


FIG. 1. Stationary Kraichnan–K41 (Gogoberidze 2007) GW spectrum $\Omega_{\rm GW}(p) \propto p^3 H_{ijij}(p,p;M,R)$, computed from Eq. (22) on the on-shell diagonal of Eq. (8), at $R = 10^4$. The five curves are $M = 0.03, 0.1, 0.3, 1, 3$, ordered by increasing amplitude and peak frequency (lowest, leftmost curve is $M = 0.03$; highest, rightmost is $M = 3$); dots mark each peak. (a) Absolute: shared IR slope p^3, amplitude growing $\sim M^3$, peak moving right with M ($p_{\rm peak} \sim M$ in the inertial range, bending over near the outer scale). (b) M^3-compensated, $\Omega_{\rm GW}/M^3$: since the IR amplitude scales as M^3 (strain as $M^{3/2}$), all curves collapse onto the universal p^3 line (dotted) in the IR and peel off only at their M-dependent peak. Generated by `Notebooks/stationary_mach_peak.py`.

D. Spectra in the $\Omega(k)/k$ convention: comparison with simulations

To compare the analytic kernel with the direct numerical simulations of Roper Pol et al. [6] we recast the spectra in their convention. There $\Omega(k)$ denotes the energy *per logarithmic wavenumber*, $\Omega(k) = d\rho/d\ln k$, so that $\Omega(k)/k = E(k)$

is the ordinary per-dk spectrum. Our master result $\Omega_{\rm GW}(p) \propto p^3 H_{ijij}(p,p)$ of Eq. (8) is precisely $d\rho_{\rm GW}/d\ln k$, hence

$$\frac{\Omega_{\rm GW}(k)}{k} \;\propto\; \frac{\Omega_{\rm GW}(p)}{p} \;=\; p^2\, H_{ijij}(p,p;M,R), \qquad p=\frac{k}{k_0}, \tag{23}$$

with H_{ijij} the stationary kernel of Eq. (22). To compare on the simulations' own footing we feed a *full* fluid input spectrum into this kernel, with both an infrared band and the Kolmogorov inertial range,

$$E(k) \;\propto\; \begin{cases} k^4, & k<k_0,\\ k^{-5/3}, & k\ge k_0, \end{cases} \tag{24}$$

joined continuously at k_0. The spectrum enters the Appendix-A integral Eq. (19) only through the amplitude $A(k) = E(k)/4\pi k^2$; writing $A(k) = S(k)\,A_{\rm Kol}(k)$ with the Kolmogorov law $A_{\rm Kol} \propto k^{-11/3}$ extrapolated to all k, the shape factor is $S(k) = 1$ for $k \ge k_0$ and $S(k) = (k/k_0)^{\,s+5/3}$ for $k < k_0$ (energy slope $E \propto k^s$, here $s = 4$). Every kinematic and temporal factor of the kernel is left untouched: we extend the k_1, u integrations down to an infrared cutoff $k_{\rm IR} = k_0/R_{\rm IR}$ and multiply the integrand by $S(k_1)\,S(u)$. Setting $R_{\rm IR} = 1$ recovers Eq. (22) identically.

Figure 2 shows the resulting GW spectrum at $M = 1$ with $R_{\rm IR} = 100$ (two infrared decades), together with the full fluid input spectrum and the two slopes the literature reports for the GW inertial range, drawn as reference guides: $k^{-11/3}$ from the simulation of [6] (run ini2) and $k^{-9/2}$ from [2]. We plot no simulation data points and defer a quantitative data-level comparison to future work. Both curves are *dimensionless*: the GW curve is the bare kernel with the source stress normalised to unity, so the GW–fluid vertical offset carries no physical meaning (the relic $\Omega_{\rm GW}$ acquires the additional suppression $\sim \Omega_{\rm M}^2\,(\mathcal{H}_*/k_*)$ and gravitational couplings that the kernel omits, which is why the simulated $\Omega_{\rm GW}/k \sim 10^{-13}$ lies far below the fluid spectrum); only slopes and peak locations are comparable, and the fluid curve is rescaled to share the GW peak.

Three features survive this honest treatment. *(i) The infrared stays causal.* The GW spectrum rises with the causal slope $\Omega_{\rm GW} \propto k^3$ ($\Omega_{\rm GW}/k \propto k^2$, fitted $k^{+2.0}$); it does *not* inherit the steeper fluid k^4 band, because the GW infrared is fixed by the analyticity of the stress rather than by the input spectrum. *(ii) The infrared band lifts the amplitude, not the inertial range.* Feeding the band raises the GW amplitude near and below the peak by a factor $\simeq 1.9$ and shifts the peak from $k \simeq 1.5\,k_0$ to $k \simeq k_0$, a change that depends only weakly on the infrared slope (a further $\simeq 1.4$ between Batchelor k^4 and Saffman k^2) and leaves the inertial range untouched. *(iii) The ultraviolet tension is robust.* Above the peak the spectrum steepens *continuously*, and at $M = 1$ far more steeply than either reference: the measured post-peak slope is $\simeq k^{-5.0}$—steeper than both Gogoberidze's $k^{-9/2}$ and the simulation's $k^{-11/3}$—before the k-dependent sweeping decorrelation bends it below any single power law. The analytic stationary kernel therefore does *not* reproduce the extended $k^{-11/3}$ range seen in the simulation; at the simulations' Mach number it is markedly steeper, crossing that slope rather than following it, and this conclusion is unchanged by the infrared treatment. This high-k disagreement is precisely the simulation–analytic tension that motivates this paper—and that the decaying kernel of Sec. III substantially relieves.

E. Discussion: the Mach number, equivalent definitions, and the spectral shape

The preceding two subsections make the Mach number the single control parameter of the stationary spectrum, so it is worth stating precisely what M is, why the conflicting definitions in the literature are the *same* quantity, and what about the spectrum does and does not respond to it.

a. Explicit formulation. Throughout we use

$$M = \left(\frac{\varepsilon}{k_0}\right)^{1/3} = u_0, \qquad (c_s \equiv 1), \tag{25}$$

the outer-scale eddy velocity in units of the sound speed (Eq. (20)). The grouping is fixed by dimensions: with $[\varepsilon] = L^2T^{-3}$ and $[k_0] = L^{-1}$ one has $\varepsilon/k_0 = (L/T)^3$, so $(\varepsilon/k_0)^{1/3}$ is a velocity and the inverse grouping $(k_0/\varepsilon)^{1/3}$ — which appears occasionally in the literature — is an *inverse* velocity and cannot be a Mach number. M does not enter Eq. (19) as a free amplitude; it enters *only* through the k-dependent sweeping rate $\eta_k = \frac{1}{\sqrt{2\pi}}\varepsilon^{1/3}k^{2/3} = \frac{M}{\sqrt{2\pi}}k_0^{1/3}k^{2/3}$, hence only through the Gaussian $\exp[-\frac{2xy}{x+y}q^2/M^2]$ and the accompanying erfc of Eq. (22).

b. Two definitions, one quantity. The literature quotes the Mach number either as a velocity ratio $M = u_{\rm rms}/c_s$ or as the cascade combination $(\varepsilon/k_0)^{1/3}$ [2]. For fully developed Kolmogorov turbulence these are the same quantity: setting $c_s = 1$ and using the inertial-range relation $\varepsilon \sim u_0^3 k_0$ turns the first into the second. The equivalence rests on three assumptions: (i) a stationary, constant-flux cascade with a well-defined ε across an inertial range of

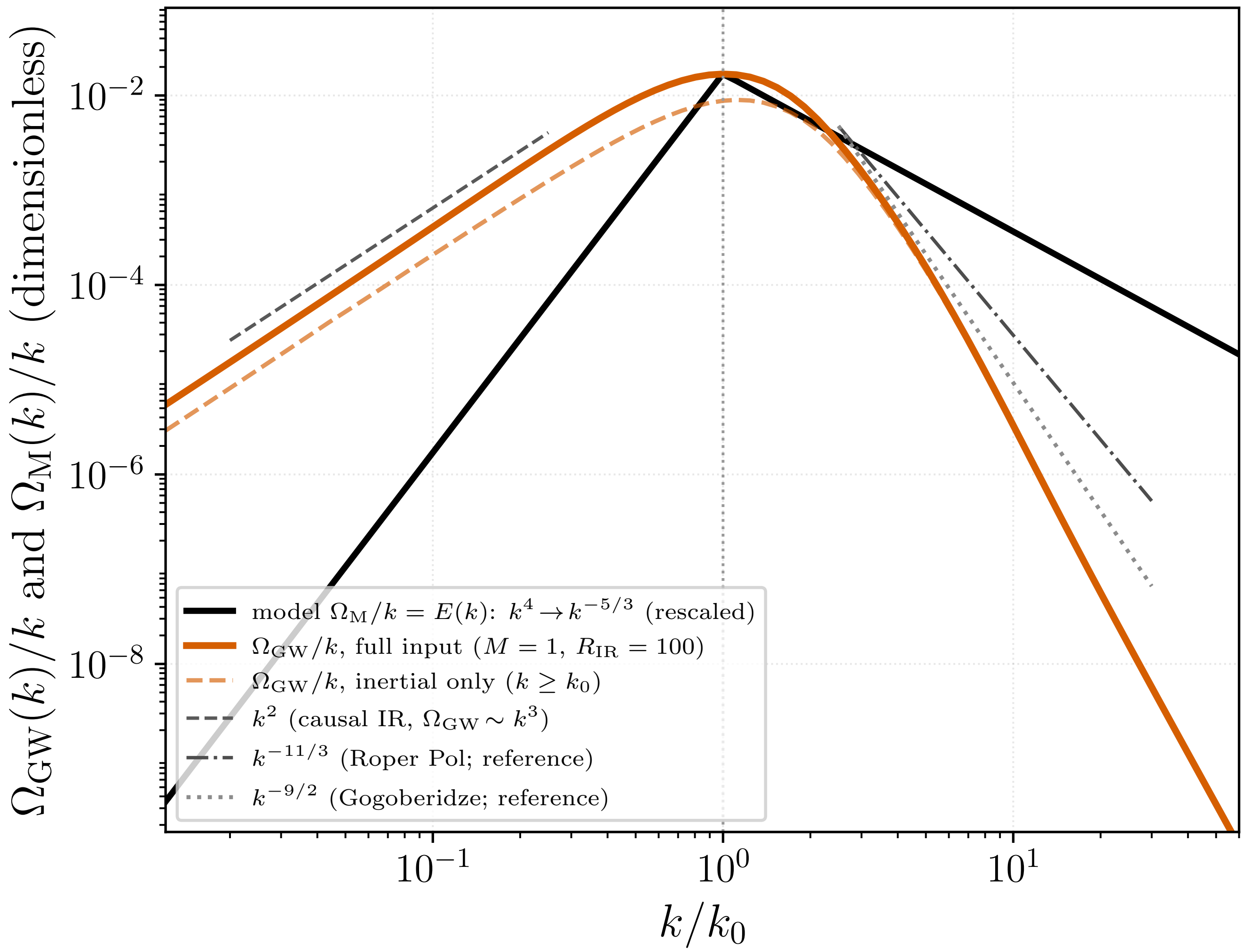


FIG. 2. Analytic spectra in the Roper Pol et al. [6] convention $\Omega(k)/k = E(k)$. Orange (solid): GW spectrum $\Omega_{\rm GW}(k)/k = p^2 H_{ijij}(p, p; M, R)$ [Eq. (23)] computed from a *full* fluid input spectrum—an infrared band $E \propto k^4$ ($k < k_0$) joined to the Kolmogorov range $E \propto k^{-5/3}$ ($k \geq k_0$), Eq. (24)—fed into the Appendix-A kernel Eq. (19) through the shape factor $S(k) = A/A_{\rm Kol}$ ($M = 1$, $R = 10^4$, infrared range $R_{\rm IR} = k_0/k_{\rm IR} = 100$). Orange (dashed): the same kernel restricted to the inertial range only ($R_{\rm IR} = 1$, i.e. core Eq. (22))—the infrared band raises the amplitude near the peak but leaves the ultraviolet unchanged. Black: the full fluid input spectrum, rescaled to share the GW peak (the GW–fluid vertical offset is unphysical; only shapes compare). Grey dashed: the causal infrared guide k^2 ($\Omega_{\rm GW} \sim k^3$). Dash-dot/dotted: the $k^{-11/3}$ (Roper Pol, run ini2) and $k^{-9/2}$ (Gogoberidze) slopes *reported* in the literature, as references only—no simulation data points are plotted. At $M = 1$ the GW spectrum steepens past the peak to $\simeq k^{-5.0}$, steeper than both references, independently of the infrared treatment. Generated by `Notebooks/roperpol_convention_spectra.py`.

width $k_d/k_0 = R^{3/4} \gg 1$; (ii) energy concentrated at the outer scale, which the $E(k) \propto k^{-5/3}$ slope guarantees because $\int_{k_0}^{\infty} E\, dk = \frac{3}{2} C_K (\varepsilon/k_0)^{2/3}$ converges at k_0; and (iii) absorbing the resulting $\mathcal{O}(1)$ constant, so that $u_{\rm rms} = \sqrt{\frac{3}{2} C_K}\, (\varepsilon/k_0)^{1/3} \approx (1.5\text{–}2)\, (\varepsilon/k_0)^{1/3}$ depending on C_K and on whether one normalises $\int E\, dk$ to $\langle u^2 \rangle$ or $\frac{1}{2}\langle u^2 \rangle$. The bare $(\varepsilon/k_0)^{1/3}$ is thus the single-eddy outer velocity, smaller than the integrated rms by this convention-dependent factor — the dominant source of numerical disagreement between papers for the same physical flow. The two definitions *diverge* when the assumptions fail: for non-Kolmogorov spectra in which the energy integral is ultraviolet-dominated (the white-noise / Batchelor model of Sec. V B, $E \propto k^2$), where $(\varepsilon/k_0)^{1/3}$ no longer measures $u_{\rm rms}$; for decaying turbulence, where $\varepsilon \to \varepsilon(t)$ and $M \to M(t)$ (Sec. VIII); and in MHD, where the relevant ratio is often the Alfvénic $M_A = u_{\rm rms}/v_A$ rather than the acoustic M [3, 7].

c. Shape versus peak versus amplitude. It is essential to separate three properties of the spectrum, only one of which is M-independent. *(i) Shape.* The dimensionless spectral form — the causal infrared slope $\Omega_{\rm GW} \propto p^3$ (Sec. I C) and the inertial power law — is universal: it is set by causality and by the source power law, not by M. This is the property that is legitimately called "M-independent," and it is what the M^{-3}-compensated collapse of Fig. 1(b) isolates. *(ii) Peak location.* Here the literature genuinely splits. In spatial-scale–anchored treatments the peak is pinned to the source scale k_0 (set by the largest eddy / bubble size), so $p_{\rm peak} = \mathcal{O}(1)$ and M controls only the amplitude [4]; in sweeping-anchored treatments the peak is tied to the eddy-turnover frequency $u_0 k_0 \propto M k_0$, so it shifts with M [2, 6]. Our kernel realises the second picture through $\eta_k \propto M k^{2/3}$, giving $p_{\rm peak} \sim M$ (Sec. II C). The two are not in contradiction: a peak "at the stirring scale" is M-independent only because it is measured in units of k_0 and the source is assumed quasi-static over an eddy. *(iii) Total energy / amplitude.* This is never M-independent. In the stationary kernel the infrared amplitude scales as the M^3 prefactor of Eq. (22) ($\Omega_{\rm GW} \to C M^3 p^3$, Sec. II C); more generally the quadrupole argument gives $\Omega_{\rm GW} \propto \Omega_K^2 \propto M^4$, steepening toward $M^{5\text{–}6}$ once the finite source duration is included [4], the extra power being absent from a single-time stationary kernel. We derive this progression explicitly for the decaying source in Sec. III B: the full-spatial kernel gives an infrared amplitude $\propto M^4$ and a source-pinned peak amplitude $\propto M^{5.3}$ [Eq. (36), Fig. 7]. Consequently a stated "M-independence" of the GW signal can only refer to the shape (i); any claim that the peak position and the total energy are simultaneously M-independent is either a normalised-units artifact (dividing by Ω_K^2 and measuring p in units of k_0) or a conflation of the universal shape with the dimensional peak and amplitude.

III. LOCATION OF THE SPECTRAL PEAK: SWEEPING VERSUS SOURCE SCALE

A striking feature of the stationary spectrum of Fig. 1 is that, for subsonic turbulence, the GW energy density peaks at wavenumbers *far below* the outer scale, $p_{\rm peak} = k_{\rm peak}/k_0 \ll 1$ (e.g. $p_{\rm peak} \simeq 0.04$ at $M = 0.03$). This is counterintuitive: the GW source is the quadratic stress $T_{ij} \sim u_i u_j$, and for a velocity field concentrated at the stirring scale k_0 one expects the stress—and hence the radiation—to peak near $k \sim k_0$, with spatial support extending to at most $k = 2k_0$ by the triangle inequality. We show here that the analytic peak is fixed instead by the *temporal* (sweeping) decorrelation, giving $p_{\rm peak} \simeq 1.47\, M$, and that the "$k \simeq 2k_0$" expectation is recovered only in the transonic limit.

a. Self-similar Mach scaling. In the dimensionless kernel of Eq. (22) the Mach number enters *only* through the sweeping rate $\eta_k \propto M k^{2/3}$, i.e. through the Gaussian $\exp[-\frac{2xy}{x+y} q^2/M^2]$ (and its accompanying erfc) evaluated on the GW cone $q = p$ of Eq. (8); the geometric kernel and the triangle-inequality bounds carry no M. Together with the explicit M^3 prefactor of Eq. (22), the on-cone kernel must therefore collapse to a single function of the sweeping variable $\xi \equiv p/M = k/(M k_0)$,

$$H_{ijij}(p, p; M) \;=\; M^3\, \widetilde{H}_{ijij}(\xi), \qquad \xi \equiv \frac{p}{M}, \tag{26}$$

where $\widetilde{H}_{ijij}$ is the *compensated kernel*: an infrared plateau $\widetilde{H}_{ijij}(0) \equiv A \simeq 2.1 \times 10^{-2}$ (the universal causal rise, Sec. I C) that rolls off once $\xi \gtrsim 1$, where the sweeping cutoff switches on. The energy density and characteristic strain follow as

$$\Omega_{\rm GW} = p^3 H_{ijij} = M^3 p^3\, \widetilde{H}_{ijij}(\xi), \qquad h_c^2 = p\, H_{ijij} = M^3 p\, \widetilde{H}_{ijij}(\xi). \tag{27}$$

b. Numerical proof of the collapse. Equation (26) is verified directly in Fig. 3(b): the compensated kernel $H_{ijij}(p, p; M)/M^3$ plotted against $\xi = p/M$ for $M = 0.03$–3 falls on the single curve $\widetilde{H}_{ijij}(\xi)$, with infrared plateau $A \simeq 2.1 \times 10^{-2}$ and $< 13\%$ residual spread dominated by the transonic $M = 3$ case. The cutoff thus depends on p and M *only* through $\xi = p/M$—the unambiguous signature of the sweeping rate—and switches on at $\xi \sim 1$, i.e. $p \sim M$. (How this single scaling is best displayed—compensating one axis or both—is taken up in Sec. III A.)

c. Peak condition and raw peak location. Differentiating $\Omega_{\rm GW} = M^3 p^3 \widetilde{H}_{ijij}(\xi)$ [Eq. (27)] on the cone,

$$\frac{d\ln \Omega_{\rm GW}}{d\ln p} \;=\; 3 + \frac{d\ln \widetilde{H}_{ijij}}{d\ln \xi} \;=\; 0 \quad\Longrightarrow\quad \left.\frac{d\ln \widetilde{H}_{ijij}}{d\ln \xi}\right|_{\xi_*} = -3, \qquad p_{\rm peak} = \xi_* M, \tag{28}$$

so the peak sits where the compensated kernel has logarithmic slope -3, at a fixed ξ_* independent of M. Reading the peak directly from each computed spectrum (Fig. 3c; sub-grid argmax, no model fit) gives a location linear in M across the entire subsonic–transonic range,

$$p_{\rm peak} \;=\; 1.48\, M^{1.00} \qquad (M \lesssim 1), \tag{29}$$

with $\xi_* = 1.47$ read from the collapsed shape, consistent with Eq. (28). Above $M \simeq 1$ the peak runs into the outer scale and the linear law saturates: $p_{\rm peak}/M$ falls from 1.49 at $M = 1$ to 0.77 at $M = 3.4$. The peak reaches the "source-scale" value $p \simeq 2$ only at transonic $M \simeq 1.4$.

d. Resolution. The naive "$k \simeq 2k_0$" is a purely *spatial* statement: it is the largest wavenumber at which the quadratic stress has support, the triangle edge $\Theta(2-p)$ that appears explicitly in the monochromatic kernel [Eq. (46)]. It is an ultraviolet *edge*, not a peak. The GW peak, by contrast, is fixed in *frequency*: gravitational waves are sourced on the cone $\omega = k$, while a turbulent eddy of wavenumber k decorrelates at the sweeping frequency $\eta_k \sim M k_0^{1/3} k^{2/3}$, which for subsonic flow is $\ll k$ for every $k \gtrsim k_0$. The source therefore has negligible temporal power at the frequency $\omega = k$ needed to radiate at wavenumber k, except at the lowest wavenumbers $k = \omega \lesssim M k_0$, and the spectrum is throttled down to $p_{\rm peak} \sim M$. This is the aeroacoustic (Lighthill) suppression of slow sources: subsonic turbulence radiates at its eddy-turnover frequency $\sim M k_0$, not at its spatial scale k_0. Whether the true peak follows this sweeping scale or remains pinned near k_0 (Sec. II E) is precisely the simulation–analytic tension this paper addresses; the stationary kernel commits unambiguously to the sweeping scale.

e. Caveat: stationarity versus the simulated peak. The scaling $p_{\rm peak} \simeq 1.47\, M$ derived above is specific to *stationary*, continuously stirred turbulence, in which the source at scale k carries temporal power only up to its eddy frequency η_k. Cosmological turbulence is not of this kind: it is sourced *impulsively*—e.g. at a phase transition—and then decays. A sudden onset of duration Δt injects broadband temporal power up to $\omega \sim 1/\Delta t \sim k_0$, which the single-time stationary kernel does not contain. Because the spatial stress is far more powerful at the outer scale k_0 than at $M k_0$, this broadband component pins the GW peak near the *spatial* stress scale, $p \sim 1$–2, almost independently of M. This is what the direct simulations of [6] display, and it is the spatial-scale picture of Sec. II E. The two predictions therefore *coincide* in the transonic band $M \sim 1$, where all current simulations sit, and *diverge* only in the subsonic corner $M \ll 1$, which no simulation has yet probed (Fig. 5). This peak displacement is the spatial counterpart of the amplitude deficit noted in Sec. II E—the stationary kernel gives $\Omega_{\rm GW} \propto M^4$ rather than the $M^{5\text{–}6}$ recovered once the finite source duration is restored—both stemming from the broadband temporal power that an impulsive, finite-duration source has and the stationary kernel omits. We compute the decaying case directly in Sec. III B below; it confirms a source-scale peak $p \simeq 2.4$, nearly independent of M, but traces it to the *shape* of the decaying temporal correlation rather than to the impulsive onset emphasised here.

A. Physical units and what the collapse hides

a. Dimensional dictionary. The kernel works in the dimensionless pair $\{p = k/k_0,\ M\}$; the observable spectrum is the characteristic strain $h_c(f)$ at the redshifted frequency f today. The two are connected by the standard cosmological map [2, 3]. A comoving mode is observed at

$$f \;=\; \frac{\hat\omega}{2\pi}\, f_H, \qquad f_H \simeq 1.6\times 10^{-5}\,{\rm Hz}\left(\frac{g_*}{100}\right)^{1/6}\frac{T_*}{100\,{\rm GeV}}, \tag{30}$$

with $\hat\omega = \omega/H_*$ and $f_H = H_*(a_*/a_0)$ the Hubble frequency today. The stirring scale in Hubble units is $\hat k_0 = 2\pi/\gamma$, where $\gamma \equiv l_0 H_*$ is the largest-eddy size in Hubble radii, so on the GW cone $\hat\omega = \hat k = \hat k_0\, p$ and

$$f \;=\; f_0\, p, \qquad f_0 \equiv \frac{f_H}{\gamma}, \tag{31}$$

i.e. $p = k/k_0$ is simply the frequency measured in units of the stirring-scale frequency f_0. Inverting, the abscissa of the standard strain figures, $(f/{\rm Hz})(g_*/100)^{-1/6}(\gamma/0.01)(T_*/100\,{\rm GeV})^{-1}$, equals p up to the fixed constant 1.6×10^{-3} and *carries no M*. For the amplitude, $\Omega_{\rm GW} = p^2 h_c^2$ with $h_c^2 = p\, H_{ijij}(p,p)$, while the physical strain (Kahniashvili Eq. 51) is $h_c = 2\times 10^{-14}(100\,{\rm GeV}/T_*)(100/g_*)^{1/3}\,(\hat\omega\,\hat\tau_T \hat H_{ijij})^{1/2}$. Multiplying h_c by $(g_*/100)^{1/3}(T_*/100\,{\rm GeV})$ cancels the cosmological prefactor, and the further factors $(\gamma/0.01)^{-3/2}(\zeta/0.01)^{-1/2}$ remove the stirring-scale and source-energy (ζ) normalisations, leaving a quantity that depends only on M and p. Both axes of the published figure are therefore our dimensionless $(p,\, h_c)$ with every *scenario* parameter $(g_*, T_*, \gamma, \zeta)$ scaled out; the single remaining knob is M.

b. Normalising the Mach dependence: one axis or two. With the scenario parameters removed, the residual Mach dependence of Eq. (27) can be scaled out in two inequivalent ways, which answer different questions.

(i) Amplitude only (y-axis). Divide the strain by $M^{3/2}$ (equivalently $\Omega_{\rm GW}$ by M^3) but keep the physical frequency $f \propto p$ on the abscissa. From Eq. (27), $h_c\, M^{-3/2} = \sqrt{p\, \widetilde H_{ijij}(p/M)}$, which is M-independent *only* in the infrared, where $\widetilde H_{ijij} \to A$: the causal rises stack, but the peaks stay at their true frequencies $f_{\rm peak} = f_0\,(1.47\, M)$ and *slide* with M. This is the convention of the published strain plots (the $h_c\, M^{-3/2}$ ordinate of [2]); it preserves the observable—where in the band each model peaks—needed to confront a detector.

(ii) Both axes. Additionally measure the frequency in sweeping units, $x \to \xi = p/M = f/(Mf_0)$. Substituting $p = M\xi$ in Eq. (27),

$$\frac{\Omega_{\rm GW}}{M^6} = \xi^3\, \widetilde{H}_{ijij}(\xi), \qquad \frac{h_c}{M^2} = \sqrt{\xi\, \widetilde{H}_{ijij}(\xi)}, \tag{32}$$

both functions of ξ *alone*: the infrared rise *and* the peak collapse onto one curve (Fig. 4). The compensation rises from M^3 to M^6 ($M^{3/2} \to M^2$ in strain) because the p^3 rise contributes an extra M^3 once the abscissa is measured in units of Mf_0 rather than f_0. This is the right diagnostic for the *theory*—it exhibits the self-similarity and proves the cutoff is a function of p/M only—but it necessarily *discards* the physical peak frequency that view (i) keeps. The two are complementary: y-only for observation, both-axis for the scaling argument.

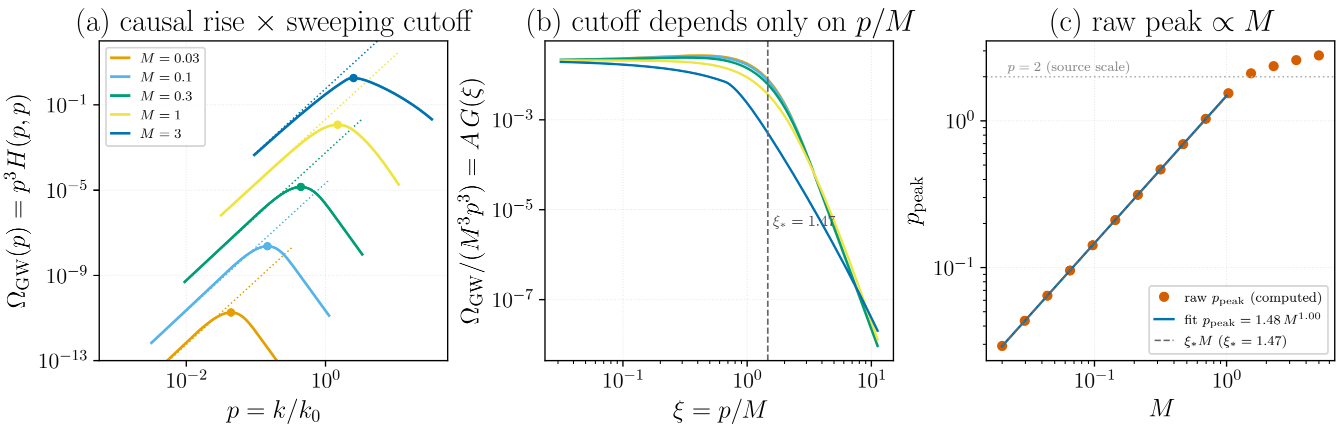


FIG. 3. Why the stationary GW spectrum peaks at $p \simeq 1.47\,M$ rather than at the source scale $p \sim 2$, computed from $\Omega_{\rm GW}(p) = p^3 H_{ijij}(p,p;M,R)$ [Eq. (22), $R = 10^4$]. (a) Each spectrum (solid) follows its causal infrared asymptote $A\,M^3p^3$ (dotted) and rolls over at the peak (dot) where the sweeping cutoff switches on. (b) Collapse: $\Omega_{\rm GW}/(M^3p^3) = H_{ijij}(p,p;M)/M^3$ versus $\xi = p/M$ falls on the single compensated kernel $\widetilde{H}_{ijij}(\xi)$ for every M, proving the cutoff depends only on p/M [Eq. (26)]; the universal peak position is $\xi_* = 1.47$. (c) Peak location read directly from each computed spectrum (points) versus M: a power law $p_{\rm peak} = 1.48\,M^{1.00}$ (line) [Eq. (29)], coincident with $\xi_* M$ (dashed), that bends below it toward the $p = 2$ source scale (dotted) only at transonic M. Generated by `Notebooks/stationary_peak_analysis.py`.

B. The decaying source pins the peak at the source scale

The caveat above is borne out quantitatively by repeating the peak analysis with the *decaying* temporal model in place of the stationary one. We keep the same full-spatial kernel—identical geometric factor and triangle-inequality integration bounds—but replace the Kraichnan Gaussian decorrelation on each stress leg by the BK2016 power law $(1 + |\tau|/\tau_k)^{-2/3}$ of Sec. V A 2, with the sweeping eddy time $\tau_k = \sqrt{2\pi}/(Mk_0^{1/3}k^{2/3})$.

a. Method: Fourier transform of the time-domain product. The temporal factor of the decaying kernel is the frequency convolution $\int d\omega_1\, \hat{g}(\omega_1\tau_{k_1})\, \hat{g}((\omega - \omega_1)\tau_u)$ of the two legs. By the convolution theorem this equals the Fourier transform of the time-domain *product* of the two decorrelations,

$$\int d\omega_1\, \hat{g}(\omega_1\tau_1)\hat{g}((\omega-\omega_1)\tau_2) \;=\; \frac{2\pi}{\tau_1\tau_2}\int_0^\infty d\tau\, \cos(\omega\tau)\left(1+\tfrac{\tau}{\tau_1}\right)^{-2/3}\left(1+\tfrac{\tau}{\tau_2}\right)^{-2/3}, \tag{33}$$

with $\tau_1 = \sqrt{x}/M$, $\tau_2 = \sqrt{y}/M$. The right-hand side, unlike the convolution—whose integrand is integrably singular at both endpoints—is smooth and rapidly convergent; evaluated this way the kernel is some two orders of magnitude cheaper to compute and well converged.

b. Result: an M-independent, source-scale peak. On the GW cone the decaying spectrum $\Omega_{\rm GW}(p) = p^3 H^{\rm decay}_{ijij}(p,p;M,R)$ peaks at

$$p^{\rm decay}_{\rm peak} \;\simeq\; 2.4, \tag{34}$$

essentially independent of both the Mach and the Reynolds number: the peak moves only from 2.29 to 2.56 as M runs over 0.3–2, and from 2.40 to 2.60 as R runs over 10^3–10^5 (a peak set by the dissipation scale would scale as $R^{3/4}$).

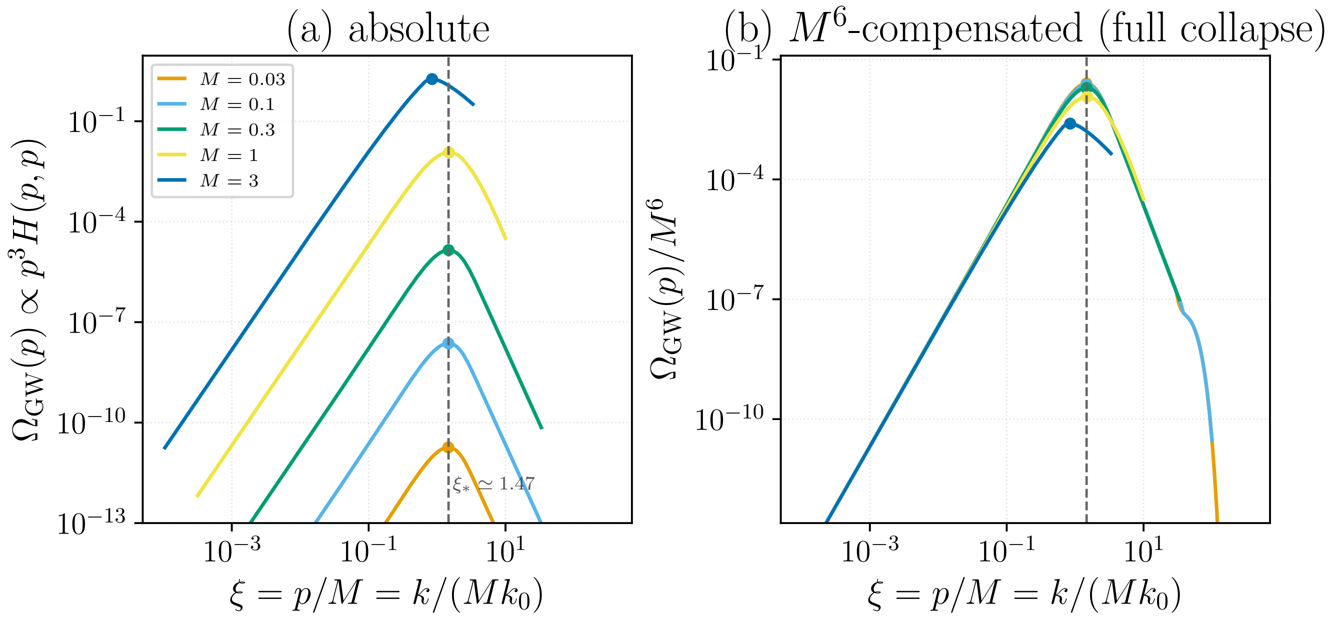


FIG. 4. Single-variable collapse of the stationary spectrum on the sweeping-scaled wavenumber $\xi = p/M = k/(Mk_0)$, computed from $\Omega_{\rm GW}(p) = p^3 H_{ijij}(p,p;M,R)$ ($R = 10^4$). (a) Absolute spectra: the peaks of all Mach numbers stack on the vertical line $\xi_* \simeq 1.47$, only their amplitudes differ ($\propto M^3$). (b) M^6-compensated spectra $\Omega_{\rm GW}/M^6$ versus ξ collapse onto the single universal curve $\xi^3 \widetilde{H}_{ijij}(\xi)$ of Eq. (32)—infrared rise and peak together—because the p^3 rise contributes an extra M^3 on the ξ axis. The subsonic curves ($M = 0.03$–1) coincide; only the transonic $M = 3$ curve (peak $\xi \simeq 0.85$) departs, where the sweeping cutoff yields to the inertial-range/triangle edge. Generated by `Notebooks/stationary_mach_peak_xM.py`.

This is in sharp contrast to the stationary law $p_{\rm peak} = 1.47\,M$ of Eq. (29), which tracks the sweeping scale. The two are compared in Fig. 6.

c. Mechanism: the high-frequency tail. The peak is fixed by the high-frequency tail of the temporal correlation. The stationary Gaussian sweeping factor $\exp[-\frac{2xy}{x+y} q^2/M^2]$ cuts the source off sharply once $q \gtrsim M$, throttling the spectrum down to $p \sim M$. The power-law decorrelation of decaying turbulence instead has a *heavy* frequency tail, $\hat{g}(q) \sim q^{-1/3}$, so the source retains power at the frequency $\omega = k$ required to radiate at every wavenumber k; the spectrum then extends past the sweeping cutoff and its peak is fixed by the spatial structure of the stress near the source scale $p \simeq 2.4$, independently of M. This realises the spatial-scale picture of Sec. II E and matches the source-scale peak of the decaying simulations [6]; the apparent contradiction with the low Gogoberidze peak dissolves once one recognises that the two correspond to opposite limits of the temporal correlation—Gaussian (rapidly decorrelating, stationary) versus power law (slowly decorrelating, decaying).

d. The decorrelation shape, not the onset. This pinning is already present in the *quasi-stationary* decaying kernel used here, in which the two-time correlation is taken at a single emission epoch and there is no impulsive switch-on. It is therefore the *shape* of the temporal decorrelation—its heavy frequency tail—rather than the broadband power of a sudden onset invoked in the caveat of Sec. III, that is responsible for displacing the peak to the source scale. Restoring the finite source duration (the self-similar slow-time integration) modifies the amplitude and overall normalisation but leaves this peak displacement intact.

e. Caveat: the peak displacement rests on a cusp that simulations dispute. This mechanism is more fragile than it looks, and the fragility is sharply localised. The heavy tail is controlled by a *single* number—the one-sided slope of the squared correlator at zero lag. For any f with $f(0)$ finite the standard Fourier result is

$$T(\omega) = 2\int_0^\infty d\tau\, \cos(\omega\tau) f(\tau) \;\longrightarrow\; -\frac{2f'(0^+)}{\omega^2}, \tag{35}$$

so the ω^{-2} tail exists if and only if the correlator has a *cusp* at $\tau = 0$. The BK2016 power law does: $f'(0^+) = -4/3$ for the squared correlation, which reproduces the coefficient $8/3$ of Eq. (50) exactly. A Gaussian does not: $f'(0^+) = 0$, and its tail then falls faster than any power.

This matters because Auclair et al. [8] find from direct hydrodynamic simulations that the unequal-time correlator of the velocity field is *Gaussian in the time difference*, $P_v \simeq P_v \exp[-\frac{1}{2}k^2 v_{\rm sw}^2(\tau-\zeta)^2]$ with $v_{\rm sw}^2 \simeq \bar{v}^2/3$, as predicted

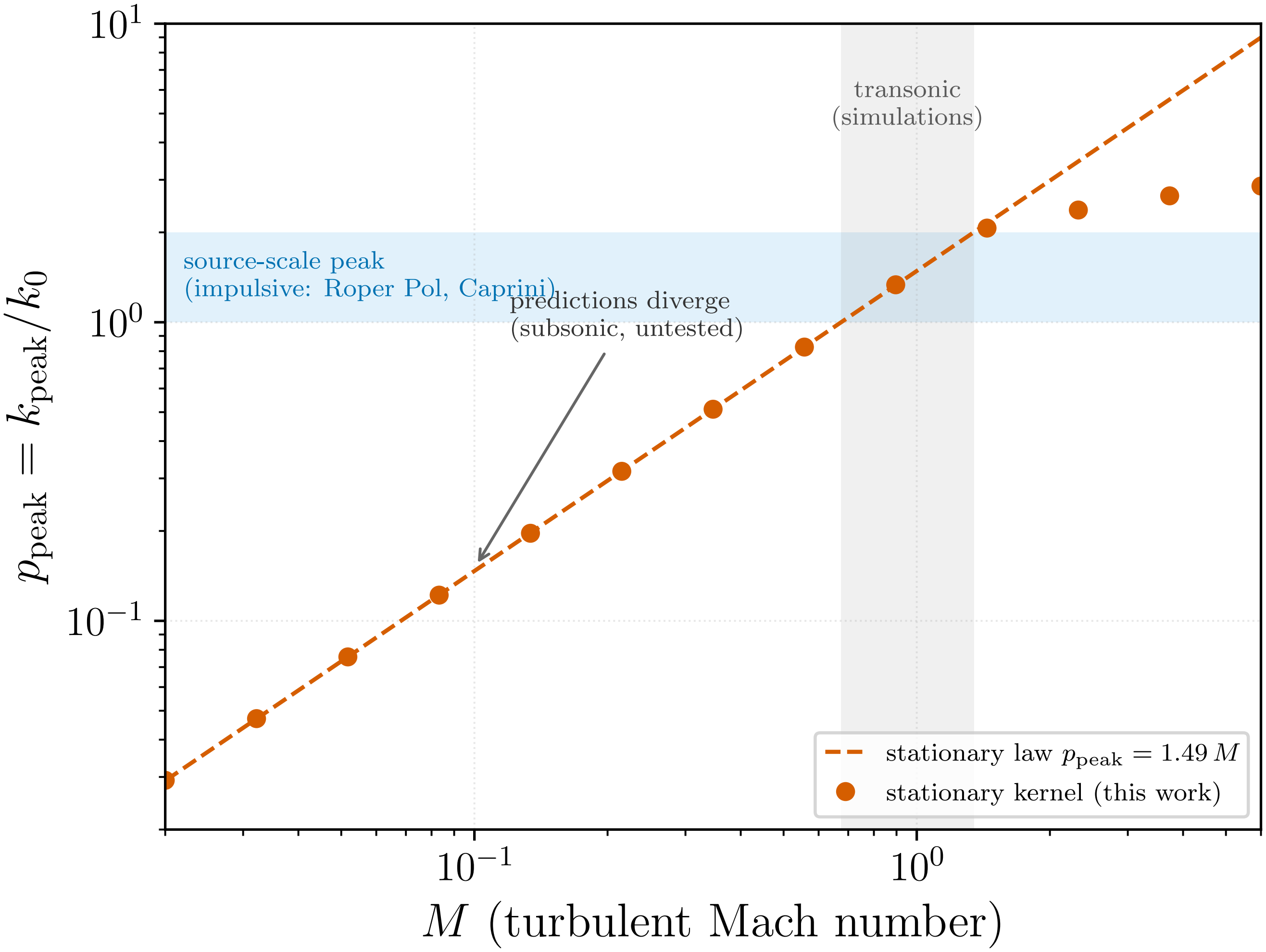


FIG. 5. Reconciliation of the stationary and impulsive pictures of the GW peak. Points and dashed line: the peak $p_{\rm peak} = k_{\rm peak}/k_0$ *computed* from the stationary kernel $\Omega_{\rm GW} = p^3 H_{ijij}(p, p; M, R)$ ($R = 10^4$), following $p_{\rm peak} \simeq 1.5\,M$. Shaded horizontal band ($p \sim 1$–2): the competing spatial-scale picture, in which an impulsively sourced, decaying flow radiates at the stress scale independently of M (Roper Pol et al. [6]; Caprini et al. [4])—a schematic, *not* computed here and not data. The two coincide in the transonic band (vertical strip), where current simulations sit, and diverge only for subsonic $M \ll 1$. Generated by `Notebooks/peak_reconciliation.py`.

by Kraichnan's random-sweeping model, and explicitly reject power-law decorrelation built on scale-dependent eddy-turnover times. Under that UETC the tail is super-exponential and the peak-pinning argument of this section does not go through; the peak would revert to the sweeping scale of Sec. III.

It should be stressed that combining the two does *not* resolve the question. One might expect a Gaussian sweeping factor to regulate the power law's cusp, but it cannot: a Gaussian has zero slope at the origin, so multiplying by it leaves $f'(0^+)$—and hence the ω^{-2} tail—unchanged (we verify slopes -1.97 to -2.13 for sweeping widths spanning a decade). The question is therefore irreducibly the one Eq. (35) poses: does the true unequal-time correlator of a freely decaying, genuinely non-stationary flow carry a $|\tau|$ kink at zero lag? The sweeping literature says no for the short-lag Eulerian correlation; whether the secular decay of the amplitude contributes one is, to our knowledge, unsettled, and the source-scale peak of Sec. III B should be read as conditional on it.

f. Amplitude: an extra power of M. The same kernel fixes the amplitude as cleanly as the peak. The Mach number enters the full-spatial decaying kernel in only two places: the prefactor $\propto M^3$ of Eq. (22) and the temporal factor Eq. (33). Rescaling $\tau \to \tau/M$ in the latter shows it equals $M\,F(\omega/M)$ with F independent of M (the geometric weight, kernel bracket and triangle bounds carry no M), so on the GW cone $\omega = k$ the spectrum factorises exactly as

$$\Omega_{\rm GW}(p; M) \;=\; M^4\, p^3\, \Psi(p,\, p/M)\,, \tag{36}$$

with Ψ the M-independent rescaled spatial integral. Two limits follow. In the infrared $p/M \to 0$ freezes $\Psi \to \Psi(p, 0)$, and the amplitude scales as the bare prefactor, $\Omega_{\rm GW} \propto M^4$ (measured: $M^{3.98}$)—one power steeper than the stationary M^3 of Eq. (22), the extra power being the single M from the temporal convolution. At the source-pinned peak $p \simeq 2.4$, by contrast, the argument p/M slides down the universal Ψ as M grows, and the scaling steepens to $\Omega_{\rm GW}^{\rm peak} \propto M^{5.3}$, with the band-integrated energy following the same $M^{5.3}$ (Fig. 7). This *derives*, rather than postulates, the $M^4 \to M^{5\text{–}6}$ progression quoted in Sec. II E: the M^4 is the genuine infrared prefactor, while the $M^{5\text{–}6}$ peak and energy steepening follows from the source-scale pinning acting on the p/M-dependent temporal weight—a mechanism already present in the quasi-stationary kernel, distinct from and prior to the finite-duration correction. (For reference the stationary peak amplitude scales similarly, $\sim M^{5.4}$, because its peak rides outward to $p = 1.47M$; the amplitude discriminant between the two pictures is therefore the *infrared* slope, M^4 versus M^3, not the peak height.)

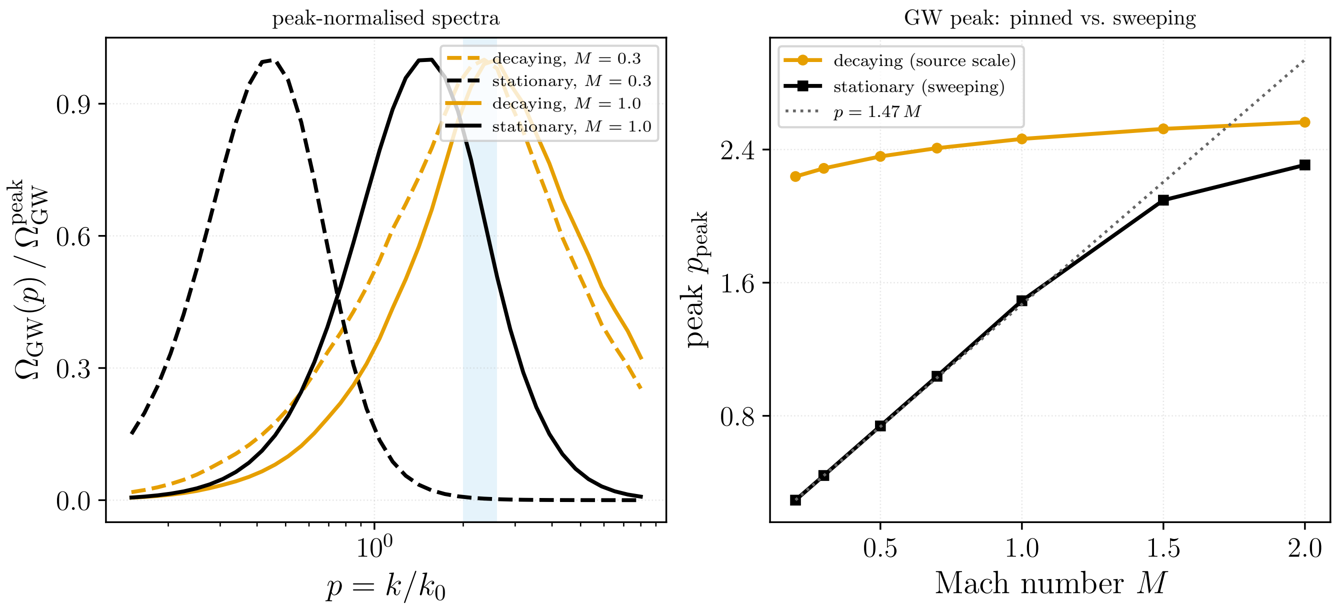


FIG. 6. The decaying source pins the GW peak at the source scale, in contrast to the stationary sweeping peak. (a) Peak-normalised spectra $\Omega_{\rm GW}(p) = p^3 H_{ijij}(p, p; M, R)$ $(R = 10^4)$ for the decaying (power-law, orange) and stationary (Gaussian, black) kernels at $M = 0.3$ (dashed) and $M = 1$ (solid): the decaying peaks fall in the source-scale band $p \simeq 2$–2.6 (shaded) for both M, whereas the stationary peaks shift with M. (b) Peak location versus M: the decaying peak is flat at $p_{\rm peak} \simeq 2.4$ [Eq. (34)], while the stationary peak follows $p_{\rm peak} = 1.47\,M$ (dotted) until it saturates near transonic M. Computed with the full-spatial kernel via the FT-of-product Eq. (33); generated by `Notebooks/fullspatial_decay.py`.

C. Confrontation with the Roper Pol simulation

We now test the two pictures against the relativistic MHD turbulence simulation of Roper Pol et al. [6] (run ini2), whose fluid and GW spectra (their Fig. 1) we digitize directly from the published figure by pixel extraction—not by eyeballed point-reading. The fluid spectrum $\Omega_{\rm M}/k$ shows the expected Batchelor infrared k^4 joined to a Kolmogorov $k^{-5/3}$ inertial range, peaking at $k_0 \simeq 673$ in the simulation's units; integrating it gives a turbulent energy fraction $\Omega_{\rm M} = \int (\Omega_{\rm M}/k)\, dk \simeq 1.8 \times 10^{-3}$, i.e. a deeply subsonic effective Mach number $M \simeq \sqrt{\Omega_{\rm M}} \sim 0.05$. Two features of the simulated GW spectrum discriminate between the sweeping and source-scale pictures.

a. Spectral shape. Figure 8 overlays our analytic $\Omega_{\rm GW}/k = p^2 H_{ijij}(p, p)$ for both temporal models on the digitized data. Each prediction's amplitude is fixed not by a free fit but by the measured energy fraction—scaled so that its total GW energy equals the simulated $\Omega_{\rm GW} = (\mathcal{H}_*/k_0)\,\Omega_{\rm M}^2$. Normalised this way the three curves coincide near the peak, and the simulated ultraviolet slope ($\simeq k^{-11/3}$, the value reported in [6]) lies *between* the two kernels: the stationary Gaussian-sweeping kernel is too steep ($\simeq k^{-5.5}$, essentially M-independent across the subsonic range) while the decaying power-law kernel is too shallow ($\simeq k^{-2.6}$). The physical decorrelation is intermediate between the rapidly-decorrelating (stationary) and slowly-decorrelating (decaying) limits; neither idealization alone reproduces the inertial-range slope.

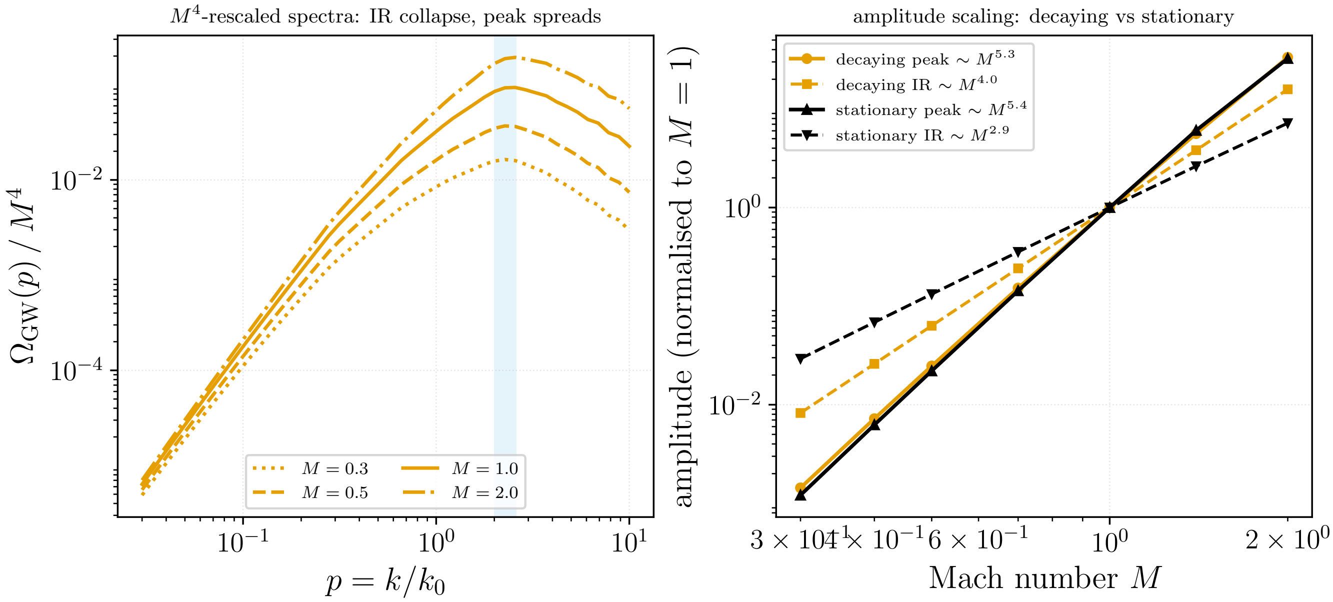


FIG. 7. Mach scaling of the full-spatial decaying GW amplitude, from the exact factorisation $\Omega_{\rm GW} = M^4 p^3 \Psi(p, p/M)$ [Eq. (36)]. (a) M^4-rescaled spectra $\Omega_{\rm GW}(p)/M^4$ for $M = 0.3$–2 ($R = 10^4$): the curves collapse in the infrared—confirming the M^4 prefactor—and spread upward through the source-scale band (shaded), the peak amplitude growing faster than M^4. (b) Amplitude versus M, normalised at $M = 1$, with fitted power laws: the decaying infrared amplitude scales as $M^{4.0}$ (one power above the stationary $M^{2.9}$), while the source-pinned peak amplitude steepens to $M^{5.3}$. Generated by `Notebooks/decaying_amplitude.py`.

b. Peak location. The sharper discriminant is the peak. In the $\Omega_{\rm GW} = d\rho_{\rm GW}/d\ln k$ convention the simulated GW peak sits at $k_{\rm peak} \simeq 1.84\,k_0$—*above* the source scale, near $2k_0$, the "twice the source scale" value expected when the quadratic stress autocorrelates a source peaked at k_0. Figure 9 plots the GW peak against M for both kernels together with this datum. The decaying kernel pins the peak at $p_{\rm peak} \simeq 2.4$ for every M and passes through the simulation point; the stationary sweeping kernel instead gives $p_{\rm peak} \simeq 1.47\,M$, which at the simulation's Mach lies at $\simeq 0.07\,k_0$, more than an order of magnitude below both the source scale and the data. This conclusion is robust to the uncertain effective Mach: across the *entire* subsonic range $M < 1$ the sweeping peak never reaches k_0, whereas the simulated peak—like the decaying kernel—lies above it. The simulated GW spectrum therefore peaks where the decaying, source-scale picture predicts, not where the stationary sweeping picture does.

IV. THE PRINCIPAL SIMULATION–ANALYTIC DISCREPANCY

Setting the two analytic treatments in the literature—the stationary Kraichnan–K41 model of Gogoberidze et al. [2] and the helical inverse-cascade model of Kahniashvili et al. [3]—beside the direct simulations of Roper Pol et al. [6] isolates one discrepancy that is both physically significant and shared by *every* analytic estimate, the kernels of this paper included. It concerns not the location of the peak (Sec. III) but the slope of the spectrum *below* the peak.

a. Three temporal pictures, three peaks. The analytic works differ in how they model the source's unequal-time correlation, and that choice fixes their spectra. Both [2] and the direct-cascade stage of [3] adopt a Gaussian decorrelation $f(\tau) = \exp(-\frac{\pi}{4}\eta_k^2\tau^2)$ with the eddy-turnover rate $\eta_k \propto \varepsilon^{1/3}k^{2/3}$, which radiates predominantly at the turnover frequency and places the GW peak near $\omega \sim Mk_0$, below the source scale for subsonic turbulence. Kahniashvili et al. add a second, dominant contribution from the helical *inverse* cascade: because magnetic helicity is conserved the energy-carrying eddy grows, and requiring its cascade time to reach the Hubble time drives the dominant peak down to the Hubble frequency $f_H \simeq 1.6 \times 10^{-5}$ Hz, essentially independent of the cascade model. The simulations [6] make no closure assumption—the source is freely decaying MHD turbulence—and find the GW peak at $k_{\rm GW} \simeq 2k_0$, twice the field scale, the elementary consequence of the source being the *quadratic* stress $T_{ij} \sim B_iB_j$ whose spectrum peaks at twice the field peak. These three pictures place the GW peak at Mk_0, at f_H, and at $2k_0$ respectively—spanning orders of magnitude (Table II); the $2k_0$ result is the one borne out in Sec. III C.

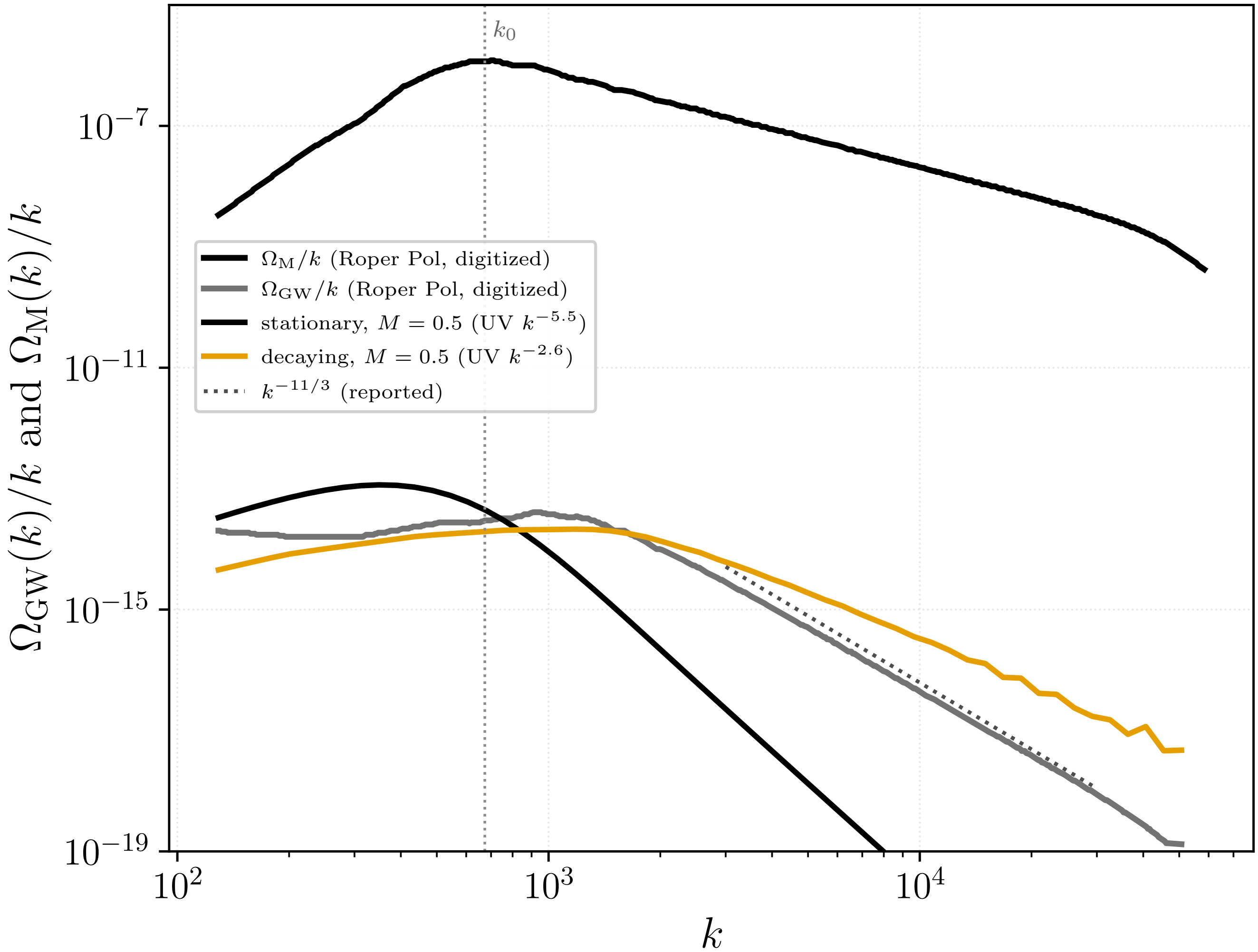


FIG. 8. Analytic GW spectra confronted with the digitized Roper Pol et al. [6] simulation (their Fig. 1, run ini2; pixel-digitized, no fabricated points). Black/grey: digitized fluid $\Omega_{\rm M}/k$ and GW $\Omega_{\rm GW}/k$. Orange/thin black: our decaying and stationary $\Omega_{\rm GW}/k = p^2 H_{ijij}(p,p)$ at $M = 0.5$ (the ultraviolet slope is M-insensitive in the subsonic range), each scaled so its total GW energy matches the simulated $\Omega_{\rm GW} = (\mathcal{H}_*/k_0)\Omega_{\rm M}^2$ (energy-fraction normalisation, no free fit). The simulated UV slope ($\sim k^{-11/3}$, dotted) falls between the too-steep stationary and too-shallow decaying kernels; the dotted vertical marks the source scale k_0. Generated by `Notebooks/roperpol_comparison.py`.

b. The discrepancy: k^3 versus k. Beneath the peak the disagreement is sharper and universal. Every *quasi-stationary* analytic treatment—the aeroacoustic Gaussian-closure framework of [2, 3] and both the stationary and decaying single-emission-epoch kernels of this paper—gives the *causal* infrared slope

$$\Omega_{\rm GW}(k) \propto k^3 \qquad (k < k_{\rm peak}), \tag{37}$$

which follows from the analyticity of the equal-time stress correlator at $k \to 0$. The simulations instead find the far shallower

$$\Omega_{\rm GW}(k) \propto k^1 \qquad (k < k_{\rm peak}), \tag{38}$$

i.e. a *flat* $\Omega_{\rm GW}(k)/k$. Roper Pol et al. single this out as their central new result—"the subinertial power law $\Omega_{\rm GW} \sim k$ is a novel result from our simulations that was not obtained in previous analytical estimates"—their Table II contrasting the analytic infrared slope 3 with the simulated slope 1 for the *same* magnetic spectrum $\Omega_{\rm M} \propto k^5$. Our own digitization of their Fig. 1 (Sec. III C) confirms it: the measured GW infrared slope is $\simeq k^{+1.3}$, not k^3 [Fig. 8].

c. Why it is significant. The discrepancy is two infrared powers of k, and it flips the sign of the strain slope: $h_c(f) \propto f^{+1/2}$ in the analytic estimates versus $f^{-1/2}$ in the simulations. Because the peak sits near the high-frequency end of the accessible band, this shallower-than-causal slope governs the *entire* sub-peak range—most of the

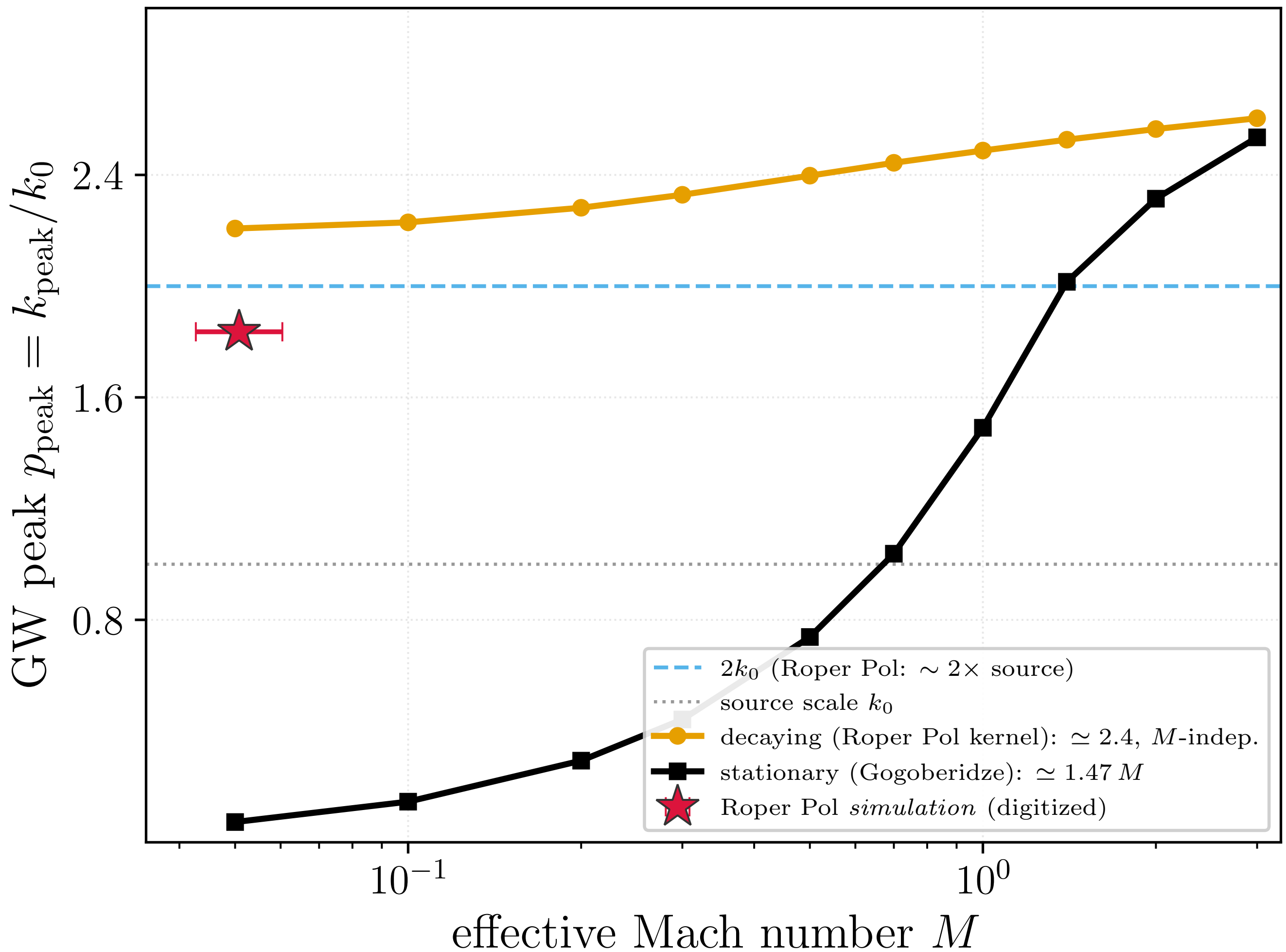


FIG. 9. GW spectral peak $k_{\rm peak}/k_0$ (in the $\Omega_{\rm GW} = d\rho/d\ln k$ convention) versus Mach number for the decaying (source-scale) and stationary (sweeping) kernels, with the digitized Roper Pol et al. [6] simulation peak (star) placed at its energy-fraction effective Mach $M \simeq 0.043$–0.060 (horizontal bar). The simulation sits at $\simeq 1.84\,k_0$, above the source scale and on the decaying branch ($\simeq 2.4$, M-independent), whereas the stationary sweeping peak $\simeq 1.47\,M$ is far below k_0 at subsonic M and never reaches it for $M < 1$. Generated by `Notebooks/gw_peak_vs_mach.py`.

observational band for an electroweak-scale source—so it changes the predicted low-frequency power, and hence the detectability, in the opposite direction to what the causal estimates assume.

d. What finite duration can and cannot do. The causal k^3 of Eq. (37) is the strict $k \to 0$ asymptote and is unavoidable: the equal-time stress of a spatially compact source is analytic (white) at $k \to 0$, and our quasi-stationary kernels reproduce it (measured infrared slope 2.93 over $p = 10^{-3}$–10^{-2}). Could the *finite duration* of the source nonetheless open a shallower band above this asymptote? Carrying the unequal-time integration over a finite emission window $[0, T_{\rm em}]$ replaces the stationary frequency factor by the finite-duration kernel

$$\int_0^{T_{\rm em}}\!\!\int_0^{T_{\rm em}} dt_1\, dt_2\; e^{i\omega(t_1-t_2)} \;\longrightarrow\; \frac{4\sin^2(\omega T_{\rm em}/2)}{\omega^2} \;\xrightarrow[\text{ens. avg}]{\omega T_{\rm em}\gg 1}\; \frac{2}{\omega^2}, \tag{39}$$

which tends to $T_{\rm em}^2$ (a constant, giving k^3) for $\omega T_{\rm em} \ll 1$ but to $2/\omega^2$ (giving k^1) for $\omega T_{\rm em} \gg 1$—*provided the source stays coherent across the window.* A source that is both coherent and sustained over $[0, T_{\rm em}]$ would thus carry a k^1 sub-peak band; this is the regime in which a source treated as statistically stationary over a finite duration reproduces the simulated slope.

e. Freely decaying turbulence does not realise it. Self-similar decaying turbulence fails the coherent-*and*-sustained condition. Its correlation time is the eddy-turnover (sweeping) time $\tau_c(k) = 1/(\varepsilon^{1/3} k^{2/3}) \propto 1/(M k^{2/3})$, fixed by the velocity and *independent* of the decay time; Kolmogorov consistency ($\varepsilon \sim u^3/L \sim u^2/\tau_{\rm st}$) ties the decay time to the outer-scale eddy turnover, $\tau_{\rm st} \sim L/u$, so the dimensionless decay rate $\varepsilon_0 = \tau_{\rm eddy}/\tau_{\rm st}$ is of order unity, not a free parameter. As the flow decays the eddy time grows, but the source amplitude falls faster ($\Phi_{\rm eq} \propto (1 + t/\tau_{\rm st})^{-34/21}$ per leg), so the eddies turn coherent only *after* they have decayed away: the growing coherence is cancelled by the decaying amplitude, and the coherent, sustained window required for k^1 is never reached for sub-peak modes. We verify this with a first-principles full-spatial self-similar kernel—the geometry of Sec. III B with the temporal factor replaced by the genuine slow-time double-time integral over $[0, T_{\rm em}]$, which reduces to the quasi-stationary kernel as $T_{\rm em} \to \infty$ (cross-check ratio $\to 1$). Its infrared slope is causal ($\simeq 2.5$, steepening to 3 as $p \to 0$) and essentially *independent of the decay rate*—2.53, 2.57, 2.51 for $\varepsilon_0 = 0$, 1, 4 [Fig. 10]. The finite duration of a decaying source does not flatten the infrared.

f. What the discrepancy points to. The causal k^3 is therefore robust for incompressible, freely decaying hydrodynamic turbulence—the stationary, decaying, and self-similar kernels of this paper all give it. The simulated k^1 of [6] must then originate in physics absent from this hydrodynamic, self-similar description, most plausibly the *magnetic* dynamics of the MHD source: magnetic-helicity conservation drives an inverse transfer that grows and sustains a coherent large-scale field, so the magnetic stress can remain coherent over the emission window in a way a decaying Kolmogorov velocity field cannot. The shallow k^1 would then be a signature of a sustained, coherent (inverse-transferred) source. On this reading the analytic–simulation tension is genuine: restoring the finite duration of a decaying source does not remove it but localises the missing ingredient in the long-time coherence of the magnetic source—a target for a dedicated MHD calculation.

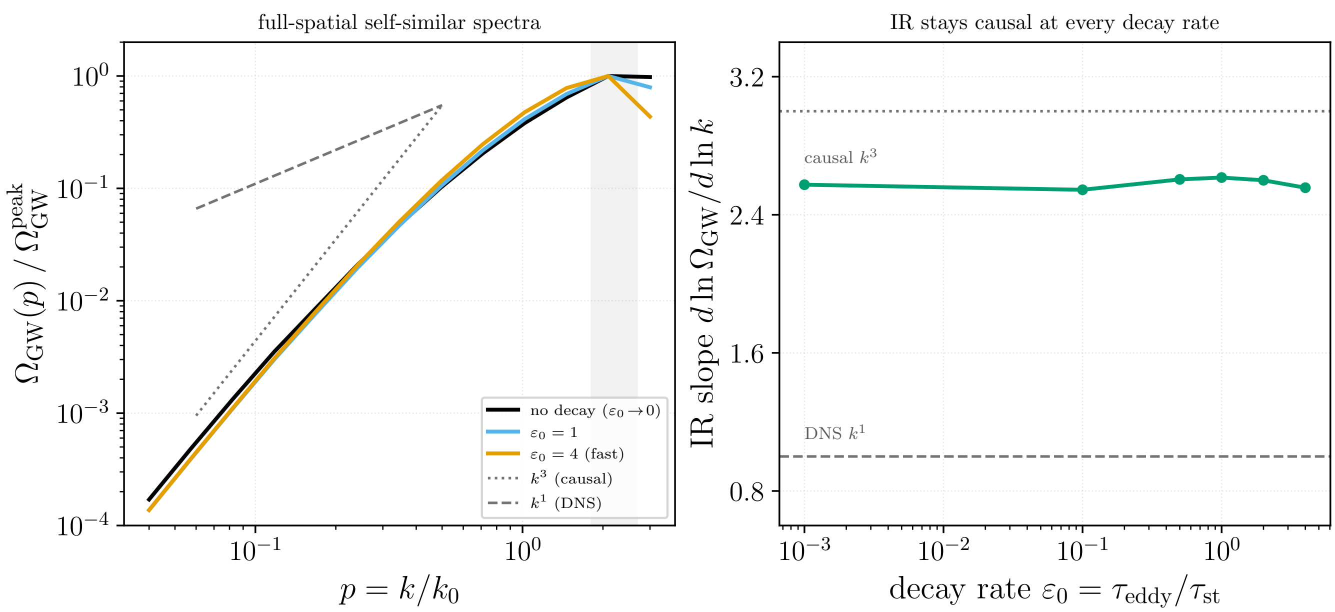


FIG. 10. The finite duration of a *decaying* source does not flatten the infrared. *Left:* GW spectra from the first-principles full-spatial self-similar kernel ($T_{\rm em} = 40\,\tau_{e0}$, $M = 1$) at three decay rates; the infrared slopes overlap on the causal k^3 guide (dotted) rather than the k^1 guide (dashed), and share the source-scale peak (shaded). *Right:* the infrared slope versus the decay rate $\varepsilon_0 = \tau_{\rm eddy}/\tau_{\rm st}$ stays at the causal value ($\simeq 2.5$–3) for every ε_0, never approaching the simulated k^1; the physically realised self-similar decay sits at $\varepsilon_0 \sim 1$. Generated by `Notebooks/fullspatial_selfsimilar_exact.py`.

A. Reproduction of the universal infrared theorem

The causal k^3 invoked repeatedly above is not a property of our kernels but a theorem. Cai, Pi and Sasaki [9] prove that $\Omega_{\rm GW} \propto k^3$ in the infrared for any source satisfying

(I) $0 < \int d\ell\,[\text{source spectra}]^2 < \infty$ (finiteness);

(II) k lies below *every* scale associated with the source—its peak k_*, its spectral width Δk, *and* the inverse duration $\Delta t_s^{-1} = (a_* \Delta\eta_s)^{-1}$;

(III) the modes re-enter the horizon during radiation domination.

treatment	temporal decorrelation	GW peak	infrared $\Omega_{\rm GW}$	high-k $\Omega_{\rm GW}$
Gogoberidze 2007 [2]	Gaussian, $\eta_k \propto k^{2/3}$	$\sim Mk_0$	k^3	steep ($\sim k^{-7/2}$)
Kahniashvili 2008 [3]	Gaussian + helical inverse cascade	$\to f_H$ (Hubble scale)	k^3	Kolmogorov input
this paper	Gaussian / power-law / self-similar decay	$1.47Mk_0$ / $2.4k_0$	k^3 (causal)	k^{-5} / $k^{-8/3}$
Roper Pol 2020 [6]	freely decaying	$2k_0$	k^1	$k^{-8/3}$

TABLE II. Temporal-decorrelation assumption, GW spectral peak, and spectral slopes for the three literature treatments and the kernels of this paper. The peak frequency spans orders of magnitude across the three works; the infrared slope is the discrepancy that is universal across *every* analytic treatment—the quasi-stationary estimates *and* the first-principles self-similar decaying kernel of this paper all give the causal k^3 (Fig. 10)—yet is contradicted by the simulation (k^1). Restoring the finite duration of a freely decaying source does *not* flatten the infrared: the growing eddy coherence is cancelled by the faster amplitude decay, so the hydrodynamic infrared stays causal for every decay rate. The tension is therefore genuine and localizes to physics absent from this hydrodynamic description (Sec. IV). The infrared slope is not quoted explicitly by [3] but is the generic causal value of their aeroacoustic Gaussian-closure framework. Slopes are in the $\Omega_{\rm GW} = d\rho_{\rm GW}/d\ln k$ convention.

The exponent itself is pure counting: at horizon crossing of $1/k_*$ there are $N = (k_*/k)^3$ causally disconnected patches, so Poisson statistics give $\langle h_k h_k\rangle = N^{-1}|h_*|^2 = (k/k_*)^3|h_*|^2$, and the k^3 follows from the dimensionality of space alone, independently of the later evolution.

This matters here because *both* of the exceptions they identify are realised by models in this paper, so the theorem is an external check rather than a restatement of our own kernels. We therefore reproduce it directly, from scratch and independently of Sec. V (Fig. 11).

a. Spectral width, and why a monochromatic source gives k^2. The anisotropic stress is the convolution of two source shells, which in one dimension carries an explicit pole,

$$\Pi(k) = \frac{2\pi}{k}\int dp\, p\, P(p)\int_{|k-p|}^{k+p} dq\, q\, P(q). \tag{40}$$

For a source of finite width the inner interval shrinks as $2k$ when $k \to 0$, cancelling the $1/k$ and leaving $\Pi(0)$ finite, so $\Omega_{\rm GW} \propto k^3$. For a δ-shell the inner integral does not shrink; the pole survives, and $\Omega_{\rm GW} \propto k^3 \cdot k^{-1} = k^2$. Numerically, a shell of log-width σ gives slope 3.000 for $k \ll \sigma$ and 2.000 in the unscreened band $\sigma \ll k \ll k_*$ (measured for $\sigma = 10^{-2}$ and 10^{-3}), while an exact δ-shell gives 2.000 over the whole infrared—a zero-width source has no lower scale, which is precisely the violation of condition (II). This is the same $1/p$ pole that appears in our monochromatic kernel as $\mathcal{K}_0(p)/p$, and it is why both panels of Fig. 14 show slope **2** rather than **3**: the monochromatic models are Cai–Pi–Sasaki's δ-function exception, not an anomaly of our temporal treatment.

b. Duration, and the intermediate k^1 band. Condition (II) also bounds k by the inverse duration. For a source coherent over a window $\Delta\eta$ the temporal factor is

$$\Big|\int_0^{\Delta\eta} d\eta\, e^{ik\eta}\Big|^2 = \frac{4\sin^2(k\Delta\eta/2)}{k^2} \;\longrightarrow\; \begin{cases}\Delta\eta^2, & k\Delta\eta \ll 1,\\ 2/k^2, & k\Delta\eta \gg 1 \text{ (oscillation-averaged)},\end{cases} \tag{41}$$

so the spectrum measures slope +2.999 below $1/\Delta\eta$ and exactly +1.000 above it: the finite emission window removes two powers. The k^1 that appears at intermediate wavenumbers is therefore not a violation of causality but condition (II) being read outside its domain—k^3 is the asymptotic law for $k \ll 1/\Delta\eta$, and k^1 is the finite-window band above it. (Cai–Pi–Sasaki quote k^9 above $1/\Delta\eta$ for acoustic sources, where a coherent *oscillating* source resonates with the Green function; for an incoherent window such as ours the loss is two powers.)

V. MONOCHROMATIC AND WHITE-NOISE SPECTRA

We now consider two alternative models for the turbulence power spectrum and carry out the same GW-kernel derivation in each case. The first model replaces the Kolmogorov spectrum by a monochromatic (delta-function) distribution in wavenumber; the second takes the real-space two-point function to be spatially uncorrelated (delta function in $\mathbf{r}$), which corresponds to a flat (white-noise) spectrum in k-space. For each spatial model we work out both the Kraichnan stationary decorrelation and the decaying-turbulence temporal model, giving four cases in total.

The derivation follows the same sequence of steps as in Sec. II: (i) write H_{ijij} as a double convolution in both $\mathbf{k}$ and ω, (ii) insert the spectral model, (iii) perform the tensor contractions to get $\mathcal{K}$, (iv) switch to spherical coordinates and the variable $u = |\mathbf{k} - \mathbf{k}_1|$, (v) evaluate or simplify the temporal integral, and (vi) introduce dimensionless variables.

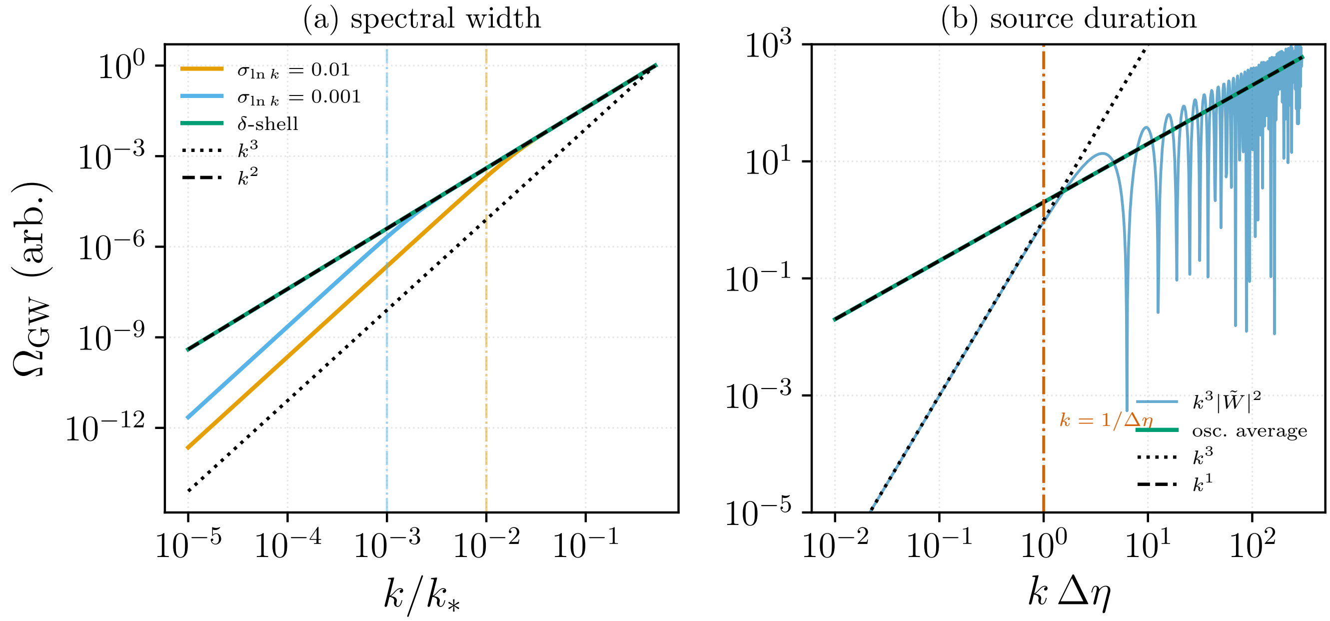


FIG. 11. Independent reproduction of the two exceptions to the universal k^3 infrared theorem of [9], computed from Eqs. (40) and (41) with no input from the kernels of this paper. *(a)* Shells of log-width $\sigma_{\ln k} = 10^{-2}$ and 10^{-3} (dash-dotted verticals mark $k = \sigma$) against an exact δ-shell: the finite-width curves follow k^3 for $k \ll \sigma$ and k^2 in the band $\sigma \ll k \ll k_*$, while the δ-shell follows k^2 throughout, having no lower scale. *(b)* Finite emission window $\Delta\eta$: the spectrum follows k^3 below $k = 1/\Delta\eta$ and, once the $\sin^2$ oscillations are averaged, k^1 above. Measured slopes are 3.000/2.000 and 2.999/1.000 respectively. Generated by `Notebooks/cai_pi_sasaki_reproduction.py`.

A. Monochromatic energy spectrum $E(k) = E_0\delta(k - k_0)$

Set $E(k) = E_0\,\delta(k - k_0)$ so that $A(k) = E_0\,\delta(k - k_0)/(4\pi k^2)$ and the spectral tensor reads

$$\tilde{F}_{ij}(\mathbf{k}, \omega) = \frac{E_0\,\delta(k - k_0)}{4\pi k^2}\,P_{ij}(\hat{k})\,\tilde{f}(\omega), \qquad \int_0^\infty E(k)\,dk = E_0, \tag{42}$$

with $\tilde{f}(\omega)$ to be specified by the decorrelation model.

Starting from Eq. (14) in spherical coordinates, the radial integral collapses $k_1 \to k_0$ via $\delta(k_1 - k_0)$, fixing $u = \sqrt{k^2 + k_0^2 - 2kk_0\mu}$. Trading μ for u (azimuth contributes 2π) and using $\delta(u - k_0)$ from $A(u)$ yields the closed-form spatial reduction

$$2\pi\int_{-1}^{1} d\mu\,A(u) = \frac{E_0}{2kk_0^2}\,\Theta(2k_0 - k), \tag{43}$$

with the Heaviside factor enforcing the triangle inequality $p \equiv k/k_0 \le 2$. The geometric kernel at $k_1 = u = k_0$ evaluates to

$$\mathcal{K}_0(p) \equiv \mathcal{K}(k, k_0, k_0) = \frac{14}{3} - \frac{p^2}{3} + \frac{p^4}{12}. \tag{44}$$

Substituting Eqs. (43)–(44) in Eq. (14),

$$H_{ijij}(\mathbf{k}, \omega) = \frac{E_0^2\,\Theta(2k_0 - k)}{2(2\pi)^8\,kk_0^2}\,\mathcal{K}_0(p)\,I_\omega(\omega), \qquad I_\omega(\omega) \equiv \int d\omega_1\,\tilde{f}(\omega_1)\tilde{f}(\omega - \omega_1). \tag{45}$$

All spatial structure has been resolved in closed form; only the temporal convolution I_ω remains. Because both wavevectors are pinned to k_0 by the two δ-functions, there is no leftover wavenumber integration: the only spatial constraint is the M-independent triangle inequality $\Theta(2 - p)$. The Mach number therefore cannot reshape any spatial integration *bounds* here; it enters solely through the temporal decorrelation rate at the fixed scale k_0, which we tie to M via $\eta_{k_0} = \frac{M}{\sqrt{2\pi}}k_0$ (cf. Eq. (20)). This contrasts with the white-noise case of Sec. V B, where M does enter the surviving wavenumber integral.

1. *Kraichnan decorrelation*

For $\tilde{f}(\omega) = (2/\eta_0)e^{-\omega^2/\pi\eta_0^2}$ with $\eta_0 \equiv \eta_{k_0} = Mk_0/\sqrt{2\pi}$, the Gaussian convolution gives $I_\omega(\omega) = (2\sqrt{2}\pi/\eta_0)e^{-\omega^2/(2\pi\eta_0^2)}$. With $q \equiv \omega/k_0$ the exponent becomes $\omega^2/(2\pi\eta_0^2) = q^2/M^2$, so Eq. (45) separates into a dimensional prefactor and the boxed dimensionless kernel

$$H_{ijij}(\mathbf{k},\omega) = \frac{\sqrt{2}\,E_0^2}{(2\pi)^7\,kk_0^2\,\eta_0}\,\mathcal{K}_0(p)\,\Theta(2-p)\,e^{-q^2/M^2}, \qquad \boxed{H[S_{\delta k},T_{\rm sw}](p,q;M) \;\equiv\; \mathfrak{H}^{(\delta k)}_{\rm Kraichnan}(p,q;M) = \frac{\mathcal{K}_0(p)}{p}\,\Theta(2-p)\,e^{-q^2/M^2}.} \tag{46}$$

There is no remaining integral. The p-shape is fixed; M acts only as the Gaussian width in q, so the GW spectrum peaks at $q \sim M$.

2. *Decay model*

For decaying turbulence, $\tilde{f}(\omega) = \hat{g}(\omega\tau_1)$ with $\hat{g}(\bar{q}) = e^{-i\bar{q}}(-i\bar{q})^{-1/3}\Gamma(\frac{1}{3},-i\bar{q})$ and $\tau_1 = 1/\eta_0 = \sqrt{2\pi}/(Mk_0)$ the decay time at scale k_0. With $q = \omega/k_0$ the natural temporal argument is $\bar{q} \equiv \omega\tau_1 = \sqrt{2\pi}\,q/M$; changing variables in Eq. (45), the temporal factor becomes $I_\omega = (1/\tau_1)\int dq_1\,\mathrm{Re}\big[\hat{g}(q_1)\hat{g}(\bar{q}-q_1)\big]$. The real part is essential and not cosmetic: the convolution is genuinely complex (its imaginary part exceeds its real part already at $\bar{q} \simeq 1$), and it is the real part that carries the spectrum. The dimensionless result is

$$\boxed{H[S_{\delta k},T_{\rm dec}](p,q;M) \;\equiv\; \mathfrak{H}^{(\delta k)}_{\rm decay}(p,q;M) = \frac{\mathcal{K}_0(p)}{p}\,\Theta(2-p)\int_{-\infty}^{\infty} dq_1\,\mathrm{Re}\big[\hat{g}(q_1)\,\hat{g}(\bar{q}-q_1)\big], \qquad \bar{q} = \frac{\sqrt{2\pi}\,q}{M}.} \tag{47}$$

The spatial part is fully resolved by the two δ-functions; only a 1-D frequency integral remains, and M enters it solely through the rescaled frequency $\bar{q}$.

a. The frequency convolution in closed form. Contrary to what the singular appearance of $\hat{g}$ suggests, this convolution *does* have a closed form, and evaluating it as a written q_1 integral is both unnecessary and numerically treacherous. The GW source is quadratic in the velocity, so under quasi-normal closure the object the spectrum needs is the transform of the *squared* two-time correlation,

$$\mathcal{C}(\bar{q}) \;\equiv\; \int_{-\infty}^{\infty} d\tau\,\cos(\bar{q}\tau)\,\big[(1+|\tau|)^{-2/3}\big]^2 \;=\; 2\,\mathrm{Re}\Big[e^{-i\bar{q}}(-i\bar{q})^{1/3}\,\Gamma\big(-\tfrac{1}{3},-i\bar{q}\big)\Big], \tag{48}$$

which is the *same* closed form as $\hat{g}$ with the power $2/3 \to 4/3$ — squaring the correlation simply shifts the exponent. Writing $h = f\Theta$ for the one-sided correlation $f(s) = (1+s)^{-2/3}$, the convolution theorem gives $(\hat{g}*\hat{g})(\bar{q}) = 2\pi\,\mathcal{F}[h^2](\bar{q})$, whose real part halves into the one-sided cosine integral. Hence the exact identity

$$\int_{-\infty}^{\infty} dq_1\,\mathrm{Re}\big[\hat{g}(q_1)\hat{g}(\bar{q}-q_1)\big] \;=\; \pi\,\mathcal{C}(\bar{q}). \tag{49}$$

Two consequences are worth stating explicitly. First, the one-sided/two-sided bookkeeping that distinguishes $\hat{g}$ from the two-sided kernel enters only as the constant π: it carries *no* shape information and therefore cannot bend a spectral slope. Second, the limits of $\mathcal{C}$ are elementary,

$$\mathcal{C}(0) = 6, \qquad \mathcal{C}(\bar{q}) \xrightarrow[\bar{q}\to 0]{} 6 - 3\sqrt{3}\,\Gamma\big(\tfrac{2}{3}\big)\,\bar{q}^{\,1/3}, \qquad \mathcal{C}(\bar{q}) \xrightarrow[\bar{q}\to\infty]{} \frac{8}{3\bar{q}^{\,2}}, \tag{50}$$

the last following from the cusp of f^2 at $\tau = 0$. This $\bar{q}^{-2}$ ultraviolet tail is the physically decisive one: it is a *heavy power law*, in contrast to the Gaussian cutoff of the stationary Kraichnan model, and it is what allows the decaying GW spectrum to extend past the sweeping cutoff so that its peak is pinned at the source scale (Sec. II E).

B. Real-space delta-function correlations

Take the spatial part of R_{ij} to be a $\delta^{(3)}(\mathbf{r})$, with velocity variance v_0^2 and temporal decorrelation $T(\eta_k,\tau)$:

$$R_{ij}(\mathbf{r},\tau) = v_0^2\,\delta^{(3)}(\mathbf{r})\,P_{ij}\,T(\eta_k,\tau), \qquad \tilde{F}_{ij}(\mathbf{k},\omega) = v_0^2\,P_{ij}(\hat{k})\,\widetilde{T}(\eta_k,\omega), \tag{51}$$

so $A(k) = v_0^2$ is k-independent (i.e. $E(k) \propto k^2$, ultraviolet-dominated); the integrals must be cut off between k_0 and k_d. Crucially, we adopt the same k-dependent sweeping rate as in Sec. II, $\eta_k = \frac{1}{\sqrt{2\pi}}\varepsilon^{1/3}k^{2/3} = \frac{M}{\sqrt{2\pi}}k_0^{1/3}k^{2/3}$, so that $\widetilde{T}$ depends on the wavenumber and the temporal factor no longer decouples from the spatial integral. With $A(k_1) = A(u) = v_0^2$ in Eq. (12) and dimensionless variables $z = k_1/k_0$, $y = u/k_0$, $p = k/k_0$, $q = \omega/k_0$, $R^{3/4} = k_d/k_0$,

$$H_{ijij}(\mathbf{k},\omega) = \frac{v_0^4 k_0^2}{(2\pi)^7}\int_1^{R^{3/4}} \frac{z\,dz}{p}\int_{\max(|p-z|,1)}^{\min(p+z,R^{3/4})} y\,dy\,\widetilde{\mathcal{K}}(p,z,y)\,J(\omega;k_1,u), \qquad J(\omega;k_1,u) \equiv \int d\omega_1\,\widetilde{T}(\eta_{k_1},\omega_1)\widetilde{T}(\eta_u,\omega-\omega_1), \tag{52}$$

where $\widetilde{\mathcal{K}}(p,z,y) = \mathcal{K}(k,k_1,u)|_{k_1=zk_0,u=yk_0}$ and the frequency factor J now carries the k_1, u dependence inherited from η_{k_1}, η_u.

1. *Kraichnan decorrelation*

With $\widetilde{T}(\eta_k,\omega) = (2/\eta_k)e^{-\omega^2/\pi\eta_k^2}$ the convolution combines two Gaussians of *different* widths η_{k_1}, η_u; using $\int d\omega_1\, e^{-a\omega_1^2}e^{-b(\omega-\omega_1)^2} = \sqrt{\pi/(a+b)}\,e^{-ab\omega^2/(a+b)}$ with $a = 1/\pi\eta_{k_1}^2$, $b = 1/\pi\eta_u^2$, and $\eta_{k_1}^2 + \eta_u^2 = \frac{M^2}{2\pi}k_0^{2/3}(k_1^{4/3} + u^{4/3})$, the frequency factor becomes

$$J(\omega;k_1,u) = \frac{4\pi\sqrt{2\pi}}{Mk_0^{1/3}\sqrt{k_1^{4/3}+u^{4/3}}}\,\exp\Bigl(-\frac{2\,\omega^2}{M^2k_0^{2/3}(k_1^{4/3}+u^{4/3})}\Bigr). \tag{53}$$

Inserting this into Eq. (52) and absorbing the constants into the dimensional prefactor gives the boxed dimensionless kernel

$$\boxed{H[S_{\delta r},T_{\rm sw}](p,q;M,R) \;\equiv\; \mathfrak{H}^{(\delta r)}_{\rm Kraichnan}(p,q;M,R) = \int_1^{R^{3/4}}\frac{z\,dz}{p}\int_{\max(|p-z|,1)}^{\min(p+z,R^{3/4})} y\,dy\,\widetilde{\mathcal{K}}(p,z,y)\,(z^{4/3}+y^{4/3})^{-1/2}\,e^{-2q^2/[M^2(z^{4/3}+y^{4/3})]}.} \tag{54}$$

The exponential now acts as an M-dependent effective cutoff on the wavenumber integration: contributions require $z^{4/3}+y^{4/3} \gtrsim 2q^2/M^2$, so as M grows the effective wavenumber support widens and the GW spectrum peak shifts. This is precisely the structure of the Gogoberidze exponent $\exp[-2\varepsilon^{-2/3}\omega^2/(k_1^{4/3}+u^{4/3})]$ in Eq. (19); the k-dependent sweeping rate, rather than a fixed η_0, is what makes M control the integration support.

2. *Decay model*

For decaying turbulence $\widetilde{T}(\eta_k,\omega) = \hat{g}(\omega\tau_k)$ with the same $\hat{g}(q) = e^{-iq}(-iq)^{-1/3}\Gamma(\frac{1}{3},-iq)$ as Sec. V A 2, but now with the k-dependent decay time $\tau_k = 1/\eta_k = \sqrt{2\pi}/(Mk_0^{1/3}k^{2/3})$. The temporal factor is the convolution $J(\omega;k_1,u) = \int d\omega_1\,\mathrm{Re}\bigl[\hat{g}(\omega_1\tau_{k_1})\hat{g}((\omega-\omega_1)\tau_u)\bigr]$ which, because τ_{k_1}, τ_u depend on the wavenumbers, must remain *inside* the spatial integral. Writing $\tilde{J}(q;z,y,M)$ for this convolution with the per-mode dimensionless frequencies fixed by $\tau_{k_1}, \tau_u \propto 1/M$, the boxed result is

$$\boxed{H[S_{\delta r},T_{\rm dec}](p,q;M,R) \;\equiv\; \mathfrak{H}^{(\delta r)}_{\rm decay}(p,q;M,R) = \int_1^{R^{3/4}}\frac{z\,dz}{p}\int_{\max(|p-z|,1)}^{\min(p+z,R^{3/4})} y\,dy\,\widetilde{\mathcal{K}}(p,z,y)\,\tilde{J}(q;z,y,M).} \tag{55}$$

The decay rates scale as $\eta_k \propto Mk_0^{1/3}k^{2/3}$, so M again sets the effective wavenumber support. Unlike the fixed-η_0 case, the three integrals no longer factorize: the k-dependence of the decay rate couples the frequency convolution to the wavenumber integration, mirroring the behavior of Eq. (19).

a. Reduction of the two-scale factor. The two-scale convolution $\tilde{J}$ admits no elementary closed form, but it reduces exactly to a one-dimensional integral over the single-scale form Eq. (48), and it is evaluated that way throughout. By the convolution theorem it is first the transform of the time-domain *product*, Eq. (33); a Feynman parameter then collapses the product of two $-2/3$ powers onto a single $-4/3$ power,

$$A^{-2/3}B^{-2/3} = \frac{\Gamma(4/3)}{\Gamma(2/3)^2}\int_0^1 \frac{dt}{[t(1-t)]^{1/3}}\,\bigl[tA+(1-t)B\bigr]^{-4/3}, \tag{56}$$

so that, with $A = 1 + \tau/\tau_1$, $B = 1 + \tau/\tau_2$ and $1/\tau_*(t) = t/\tau_1 + (1-t)/\tau_2$,

$$\int_{-\infty}^{\infty} d\omega_1 \,\mathrm{Re}\big[\hat{g}(\omega_1\tau_1)\hat{g}((\omega-\omega_1)\tau_2)\big] = \frac{\pi}{\tau_1\tau_2}\,\frac{\Gamma(4/3)}{\Gamma(2/3)^2}\int_0^1 \frac{dt}{[t(1-t)]^{1/3}}\,\tau_*(t)\,\mathcal{C}\big(\omega\,\tau_*(t)\big). \tag{57}$$

The t-integrand is smooth and non-oscillatory ($\mathcal{C}$ is positive and monotonically decreasing), and its endpoint singularities are exactly the Gauss–Jacobi weight, so a fixed 48-node rule is converged to ten digits; at $\tau_1 = \tau_2$ it collapses back to Eq. (49), since $[\Gamma(4/3)/\Gamma(2/3)^2]\,B(2/3, 2/3) = 1$. This matters numerically as well as formally: evaluated as a written oscillatory integral the amplitude varies on τ_1, τ_2 while the cosine varies on $1/\omega$, and at small ω those scales are separated by many decades, so standard oscillatory quadratures either fail outright or return silently incorrect values. Eq. (57) contains no oscillatory quadrature at all.

Summary of the four dimensionless integrals

Spectrum	Decorrelation	Dimensionless integral
$E_0\delta(k-k_0)$	Kraichnan	Eq. (46): closed form; M in frequency scale only
$E_0\delta(k-k_0)$	Decay	Eq. (47): 1D frequency integral; M in $\bar{q}$
$\delta^{(3)}(\mathbf{r})$	Kraichnan	Eq. (54): 2D wavevector integral; M-cutoff in bounds
$\delta^{(3)}(\mathbf{r})$	Decay	Eq. (55): 2D wavevector $\times$ nested 1D frequency; M-cutoff
$E_0\delta(k-k_0)$	Impulsive $\delta(\tau-\tau_s)$	Eq. (70): closed form; no M or q dependence

VI. IMPULSIVE (δ-IN-TIME) SOURCE

We now consider a source that fires exactly once: the turbulent stress is active only at a single conformal time $\tau = \tau_s$ but remains monochromatic in wavenumber, $E(k) = E_0\,\delta(k-k_0)$. This idealisation approximates a combustion front, a rapid phase transition, or any process whose active lifetime is negligible compared with the GW oscillation period $2\pi/k$. Because the temporal part of the UETC is an exact double δ-function, the time integrals in the master formula collapse *without approximation*, yielding a GW spectrum that is purely spatial in character and entirely free of Mach-number or decorrelation-time parameters.

A. Source model and UETC

The stress tensor of the impulsive source factorises as

$$\Pi_{ij}^{\mathrm{TT}}(\mathbf{k},\tau) = A_{ij}^{\mathrm{TT}}(\mathbf{k})\,\delta(\tau-\tau_s), \tag{58}$$

where $A_{ij}^{\mathrm{TT}}(\mathbf{k})$ carries the spatial amplitude of the burst and is independent of time. The two-point function of the TT stress defines the UETC via

$$\big\langle \Pi_{ij}^{\mathrm{TT}}(\mathbf{k},\tau_1)\,\Pi_{ij}^{\mathrm{TT}\,*}(\mathbf{k}',\tau_2)\big\rangle = (2\pi)^3\,\delta^{(3)}(\mathbf{k}-\mathbf{k}')\,\Pi(k;\tau_1,\tau_2). \tag{59}$$

Inserting the ansatz (58) and defining the spatial power $P(k)$ through $\langle A_{ij}^{\mathrm{TT}}(\mathbf{k})\,A_{ij}^{\mathrm{TT}\,*}(\mathbf{k}')\rangle = (2\pi)^3\delta^{(3)}(\mathbf{k}-\mathbf{k}')P(k)$, the UETC becomes

$$\Pi(k;\tau_1,\tau_2) = P(k)\,\delta(\tau_1-\tau_s)\,\delta(\tau_2-\tau_s). \tag{60}$$

No decorrelation model (η_k, $\hat{g}$, or any finite coherence time) is needed; the UETC is exact and requires no further closure assumption.

B. Temporal integral and master formula

The Isaacson GW energy density $\rho_{\mathrm{GW}} = \langle \dot{h}_{ij}\dot{h}_{ij}\rangle/(32\pi G)$, together with the retarded solution

$$\dot{h}_{ij}(\mathbf{k},\tau) = 16\pi G\int_{\tau_i}^{\tau} d\tau'\,\cos[k(\tau-\tau')]\,\Pi_{ij}^{\mathrm{TT}}(\mathbf{k},\tau'), \tag{61}$$

and time-averaging the fast-oscillation product $\cos[k(\tau-\tau_1)]\cos[k(\tau-\tau_2)] \to \frac{1}{2}\cos[k(\tau_1-\tau_2)]$, gives the master formula

$$\frac{d\rho_{\rm GW}}{d\ln k} \simeq \frac{2G\,k^3}{\pi}\int d\tau_1\int d\tau_2\,\cos[k(\tau_1-\tau_2)]\,\Pi(k;\tau_1,\tau_2). \tag{62}$$

Substituting the impulsive UETC (60),

$$\begin{aligned}\frac{d\rho_{\rm GW}}{d\ln k} &= \frac{2G\,k^3}{\pi}\int d\tau_1\int d\tau_2\,\cos[k(\tau_1-\tau_2)]\;P(k)\,\delta(\tau_1-\tau_s)\,\delta(\tau_2-\tau_s)\\ &= \frac{2G\,k^3}{\pi}\,P(k)\,\cos[k(\tau_s-\tau_s)] \;=\; \frac{2G\,k^3}{\pi}\,P(k)\,\cos(0) \;=\; \frac{2G\,k^3}{\pi}\,P(k).\end{aligned} \tag{63}$$

The temporal factor is identically unity regardless of k, k_0, or τ_s: both δ-functions collapse to set $\tau_1=\tau_2=\tau_s$, so the cosine argument vanishes. No approximation is involved. The GW spectrum is therefore determined entirely by the instantaneous spatial power $P(k)$ of the burst, with no temporal modulation and no on-shell frequency integral.

C. Spatial reduction for monochromatic $E(k) = E_0\,\delta(k-k_0)$

We now compute $P(k)$ for the monochromatic energy spectrum. Adopting the isotropic, incompressible velocity model $F_{ij}(\mathbf{k}) = A(k)P_{ij}(\hat{k})$ with $P_{ij}(\hat{k}) = \delta_{ij}-\hat{k}_i\hat{k}_j$, the spatial power of the TT-projected quadratic source reads

$$P(k) = \frac{1}{(2\pi)^7}\int d^3k_1\,A(k_1)\,A(u)\,\mathcal{K}(k,k_1,u), \qquad \mathbf{u}\equiv\mathbf{k}-\mathbf{k}_1, \quad u\equiv|\mathbf{u}|, \tag{64}$$

with the geometric kernel

$$\mathcal{K}(k,k_1,u) = \frac{27}{6}+\frac{k^4}{12k_1^2u^2}+\frac{k_1^2}{12u^2}+\frac{u^2}{12k_1^2}-\frac{k^2}{6u^2}-\frac{k^2}{6k_1^2}. \tag{65}$$

For the monochromatic spectrum $E(k) = E_0\,\delta(k-k_0)$, the amplitude is $A(k) = E_0\,\delta(k-k_0)/(4\pi k^2)$. Going to spherical coordinates $d^3k_1 = k_1^2\,dk_1\,d\phi\,d\mu$ with $\mu = \hat{k}\cdot\hat{k}_1$, the azimuthal integral gives 2π; trading μ for $u = \sqrt{k^2+k_1^2-2kk_1\mu}$ (so $u\,du = kk_1\,d\mu$),

$$P(k) = \frac{1}{(2\pi)^6}\int_0^\infty \frac{k_1\,dk_1}{k}\int_{|k-k_1|}^{k+k_1} u\,du\;A(k_1)\,A(u)\,\mathcal{K}(k,k_1,u). \tag{66}$$

Inserting $A(k_1) = E_0\delta(k_1-k_0)/(4\pi k_1^2)$, the k_1 integral collapses to $k_1=k_0$; inserting $A(u) = E_0\delta(u-k_0)/(4\pi u^2)$, the u integral collapses to $u=k_0$, which is inside the range $[|k-k_0|, k+k_0]$ only when $k\le 2k_0$, i.e. $p\le 2$. The angular residue from $\delta(u-k_0)$ is $1/(du/d\mu)|_{u=k_0} = u/(kk_1)|_{u=k_0} = k_0/(kk_0) = 1/k$, so the two-dimensional integration reduces to

$$\int_0^\infty \frac{k_1\,dk_1}{k}\int_{|k-k_1|}^{k+k_1} u\,du\;\frac{E_0\,\delta(k_1-k_0)}{4\pi k_1^2}\cdot\frac{E_0\,\delta(u-k_0)}{4\pi u^2} = \frac{E_0^2}{16\pi^2}\cdot\frac{1}{k_0^2}\cdot\frac{1}{k}\,\Theta(2k_0-k). \tag{67}$$

With both momenta pinned at k_0, the geometric kernel evaluates to

$$\begin{aligned}\mathcal{K}(k,k_0,k_0) &= \frac{27}{6}+\frac{k^4}{12k_0^4}+\frac{k_0^2}{12k_0^2}+\frac{k_0^2}{12k_0^2}-\frac{k^2}{6k_0^2}-\frac{k^2}{6k_0^2}\\ &= \frac{27}{6}+\frac{p^4}{12}+\frac{1}{12}+\frac{1}{12}-\frac{p^2}{6}-\frac{p^2}{6}\\ &= \frac{27+1+1}{6}+\frac{p^4}{12}-\frac{p^2}{3} \;=\; \frac{14}{3}-\frac{p^2}{3}+\frac{p^4}{12} \;\equiv\; \mathcal{K}_0(p).\end{aligned} \tag{68}$$

Substituting (67) and (68) into (66) gives

$$P(k) = \frac{E_0^2\,\mathcal{K}_0(p)}{(2\pi)^6\cdot 16\pi^2\,k\,k_0^2}\,\Theta(2-p). \tag{69}$$

Inserting into (63) and collecting the dimensional prefactor into an overall constant, we read off the dimensionless kernel

$$\boxed{\mathfrak{H}^{(\delta\tau,\delta k)}(p) = \frac{\mathcal{K}_0(p)}{p}\,\Theta(2-p), \qquad \mathcal{K}_0(p) = \frac{14}{3} - \frac{p^2}{3} + \frac{p^4}{12}.} \tag{70}$$

There is no remaining integral, no frequency argument q, and no Mach number M: the kernel is a single algebraic expression in p. In p-shape it coincides with the Kraichnan monochromatic kernel of Sec. II evaluated at $q = 0$ — setting $q \to 0$ in e^{-q^2/M^2} trivially reproduces (70) — because the impulsive source assigns equal weight to every on-shell frequency.

D. GW spectrum: peak, IR and UV slopes

Using $\Omega_{\rm GW}(p) \propto p^3\,\mathfrak{H}^{(\delta\tau,\delta k)}(p)$,

$$\Omega_{\rm GW}(p) \;\propto\; p^2\,\mathcal{K}_0(p)\,\Theta(2-p). \tag{71}$$

a. Infrared slope. As $p \to 0$, $\mathcal{K}_0(p) = \frac{14}{3} + O(p^2)$ is constant, so

$$\Omega_{\rm GW}(p) \xrightarrow{p\to 0} \frac{14}{3}\,p^2. \tag{72}$$

The infrared slope is p^2, not the causal p^3 of spatially extended sources. The extra power of $1/p$ in $\mathfrak{H}$ traces back to the angular residue in (67): pinning both loop momenta to k_0 means the phase-space measure for small k grows as k rather than k^2, which demotes $k^3 \to k^2$.

b. Peak. Define $f(p) \equiv p^2\,\mathcal{K}_0(p) = \frac{14}{3}\,p^2 - \frac{p^4}{3} + \frac{p^6}{12}$. Its derivative is

$$f'(p) = p\left(\frac{28}{3} - \frac{4}{3}\,p^2 + \frac{p^4}{2}\right). \tag{73}$$

Setting the bracket to zero requires $3p^4 - 8p^2 + 56 = 0$, a quadratic in p^2 with discriminant $\Delta = 64 - 4\cdot 3\cdot 56 = -608 < 0$: no real roots. Since the bracket is $56/3 > 0$ at $p = 0$ and $72 > 0$ at $p = 2$, and has no roots on $(0, 2)$, $f'(p) > 0$ throughout the support. The spectrum is strictly increasing and peaks at the triangle-inequality cutoff,

$$p_{\rm peak} = 2 \qquad \Longrightarrow \qquad k_{\rm GW}^{\rm peak} = 2k_0. \tag{74}$$

Geometrically, the maximum GW wavenumber from two modes both at k_0 is $k_0 + k_0 = 2k_0$; the monotone kernel drives all power to this kinematic limit.

c. Ultraviolet. The Heaviside factor $\Theta(2-p)$ enforces an exact hard cutoff: $\Omega_{\rm GW}(p) \equiv 0$ for $p > 2$. There is no UV power-law tail, in contrast to the smooth Gaussian (Kraichnan) or power-law (decay) falloffs of the stationary models.

E. Post-burst free propagation

For $\tau > \tau_s$ the source is off. The retarded Green's function gives

$$h_{ij}(\mathbf{k},\tau) = \frac{16\pi G}{k}\int_{\tau_i}^{\tau} d\tau'\,\sin[k(\tau-\tau')]\,\Pi_{ij}^{\rm TT}(\mathbf{k},\tau'). \tag{75}$$

Substituting $\Pi_{ij}^{\rm TT}(\mathbf{k},\tau') = A_{ij}^{\rm TT}(\mathbf{k})\,\delta(\tau'-\tau_s)$,

$$h_{ij}(\mathbf{k},\tau) = \frac{16\pi G}{k}\,\sin[k(\tau-\tau_s)]\,A_{ij}^{\rm TT}(\mathbf{k}), \qquad \tau > \tau_s, \tag{76}$$

with time derivative $\dot{h}_{ij}(\mathbf{k},\tau) = 16\pi G\cos[k(\tau-\tau_s)]\,A_{ij}^{\rm TT}(\mathbf{k})$. The Isaacson energy density

$$\rho_{\rm GW} = \frac{\langle \dot{h}_{ij}\dot{h}_{ij}\rangle}{32\pi G} = \frac{(16\pi G)^2}{32\pi G}\cos^2[k(\tau-\tau_s)]\,P(k) \xrightarrow{\langle\cos^2\rangle = 1/2} \frac{(16\pi G)\,k^3}{2\pi}\,P(k)\,\frac{d\ln k}{1}, \tag{77}$$

after integrating over k-shells, recovers the frozen spectrum (71) without further time evolution. Each mode oscillates on-shell at $\omega = k$ with amplitude set at τ_s; the GW background decouples from the source permanently and carries a spectrum that never changes.

VII. IMPULSIVE SOURCE IN REAL SPACE

We keep every space integral in configuration space and transform in time alone. The GW spectrum comes out as a single frequency ω acting on a purely spatial stress correlator. With $a = 1$,

$$\Box\, h_{ij} = 16\pi G\, S_{ij}, \quad \Box = \partial_t^2 - \nabla^2; \qquad \dot{h}_{ij}(\mathbf{r}, t) = 4G \int d^3r'\, \frac{\dot{S}_{ij}(\mathbf{r}', t - |\mathbf{r} - \mathbf{r}'|)}{|\mathbf{r} - \mathbf{r}'|}. \tag{78}$$

Following the standard turbulence–GW formalism [1, 2], introduce the unequal-time GW correlator at lag τ and its temporal transform,

$$L(\mathbf{r}, \tau) \equiv \frac{1}{32\pi G} \big\langle \dot{h}_{ij}(\mathbf{r}, t)\, \dot{h}_{ij}(\mathbf{r}, t + \tau) \big\rangle, \qquad \rho_{\rm GW}(\mathbf{r}) = L(\mathbf{r}, 0), \tag{79}$$

$$I(\mathbf{r}, \omega) \equiv \frac{1}{2\pi} \int d\tau\, e^{i\omega\tau}\, L(\mathbf{r}, \tau), \qquad \rho_{\rm GW}(\mathbf{r}) = \int d\omega\, I(\mathbf{r}, \omega), \tag{80}$$

so $I(\mathbf{r}, \omega)$ is the spectral energy density of the radiation and $d\rho_{\rm GW}/d\ln\omega = \omega\, I(\mathbf{r}, \omega)$ (at $\omega > 0$). Inserting the retarded field (78) twice ($(4G)^2/32\pi G = G/2\pi$),

$$L(\mathbf{r}, \tau) = \frac{G}{2\pi} \int \frac{d^3r_1\, d^3r_2}{|\mathbf{r} - \mathbf{r}_1|\, |\mathbf{r} - \mathbf{r}_2|} \big\langle \dot{S}_{ij}(\mathbf{r}_1, t')\, \dot{S}_{ij}(\mathbf{r}_2, t'') \big\rangle, \quad t' = t - |\mathbf{r} - \mathbf{r}_1|, \quad t'' = t + \tau - |\mathbf{r} - \mathbf{r}_2|. \tag{81}$$

For turbulence the stress obeys the identity [2]

$$\big\langle \dot{S}_{ij}(\mathbf{r}_1, t')\, \dot{S}_{ij}(\mathbf{r}_2, t'') \big\rangle = -\partial_\tau^2 \big\langle S_{ij}(\mathbf{r}_1, t')\, S_{ij}(\mathbf{r}_2, t'') \big\rangle, \tag{82}$$

so under the transform (80) the two time-derivatives integrate by parts in τ to a factor ω^2,

$$I(\mathbf{r}, \omega) = \frac{G\,\omega^2}{(2\pi)^2} \int \frac{d^3r_1\, d^3r_2}{|\mathbf{r} - \mathbf{r}_1|\, |\mathbf{r} - \mathbf{r}_2|} \int d\tau\, e^{i\omega\tau} \big\langle S_{ij}(\mathbf{r}_1, t')\, S_{ij}(\mathbf{r}_2, t'') \big\rangle. \tag{83}$$

A single burst $S_{ij} = \mathcal{S}_{ij}(\mathbf{r})\,\delta(t - t_0)$ gives $\langle S_{ij}(\mathbf{r}_1, t') S_{ij}(\mathbf{r}_2, t'')\rangle = \Sigma(\mathbf{r}_1, \mathbf{r}_2)\,\delta(t' - t_0)\,\delta(t'' - t_0)$ with $\Sigma(\mathbf{r}_1, \mathbf{r}_2) = \langle \mathcal{S}_{ij}(\mathbf{r}_1) \mathcal{S}_{ij}(\mathbf{r}_2)\rangle$; the δ's fix the retarded shells (so t_0 sets only the shell radius, not the spectral shape), the τ-integral leaves the phase $e^{-i\omega(|\mathbf{r}-\mathbf{r}_1| - |\mathbf{r}-\mathbf{r}_2|)}$, and with $\mathbf{r} = 0$ ($|\mathbf{r} - \mathbf{r}_a| = r_a$), the isotropic correlator $\Sigma(r)$ ($r = |\mathbf{r}_1 - \mathbf{r}_2|$), and the angular average $\langle e^{i\boldsymbol{\omega}\cdot\mathbf{r}}\rangle = \sin(\omega r)/(\omega r)$,

$$\boxed{\frac{d\rho_{\rm GW}}{d\ln\omega} = 8G\,\omega^2 \int_0^\infty dr\, r\, \sin(\omega r)\, \Sigma(r).} \tag{84}$$

The leading ω^2 is exactly the two time-derivatives of the quadrupole source, converted by the identity (82).

A. The peak is set by the source, not by t_0

The firing time enters (83) only through the δ's, which fix the radiating shell but leave no t_0 in the real spectrum (84): *the spectrum is independent of* t_0. An instantaneous burst nonetheless radiates a broadband, structured spectrum, fixed entirely by the spatial stress correlator $\langle \mathcal{S}_{ij}(\mathbf{r}_1) \mathcal{S}_{ij}(\mathbf{r}_2)\rangle$. Let ℓ be its coherence length (the correlator decays for $|\mathbf{r}_1 - \mathbf{r}_2| \gtrsim \ell$). The retardation phase $\cos[\omega(r_1 - r_2)]$ stays coherent across that volume while $\omega\ell \lesssim 1$ and averages away above it, so the radial integral is white in the infrared and the causal rise $d\rho_{\rm GW}/d\ln\omega \propto \omega^3$ turns over at

$$\boxed{\omega_{\rm peak} \simeq \frac{1}{\ell} \quad (\text{independent of } t_0), \qquad f_{\rm peak} = \frac{\omega_{\rm peak}}{2\pi} \simeq \frac{1}{2\pi\ell}.} \tag{85}$$

The peak GW frequency is the inverse coherence length of the burst, with no reference to when it fired.

B. Burst of finite duration Δt

Now let the stress switch on over a finite interval rather than an instant, $S_{ij}(\mathbf{r},t) = \mathcal{S}_{ij}(\mathbf{r})\,g(t-t_0)$, with g a temporal profile of width Δt (the instantaneous burst is $\Delta t \to 0$, $g \to \delta$). The source correlator in (83) now carries $g(t'-t_0)\,g(t''-t_0)$ instead of the two δ's, so the τ-transform multiplies the spectrum by the single temporal filter $|\tilde g(\omega)|^2$, $\tilde g(\omega) = \int dt\, g(t) e^{i\omega t}$,

$$\boxed{\frac{d\rho_{\rm GW}}{d\ln\omega} = 8G\,\omega^2\,|\tilde g(\omega)|^2 \int_0^\infty dr\, r\,\sin(\omega r)\,\Sigma(r).} \tag{86}$$

The factor $|\tilde g(\omega)|^2$ is flat for $\omega\Delta t \lesssim 1$ and rolls off above — a low-pass at $\omega \sim 1/\Delta t$. The spectrum now carries two scales, the spatial coherence ($\omega \sim 1/\ell$) and the duration ($\omega \sim 1/\Delta t$), and peaks at the lower of the two,

$$\boxed{\omega_{\rm peak} \simeq \min\!\Big(\frac{1}{\ell}, \frac{1}{\Delta t}\Big).} \tag{87}$$

A slow burst ($\Delta t \gtrsim \ell$) is duration-limited, $\omega_{\rm peak} \simeq 1/\Delta t$; a fast burst ($\Delta t \lesssim \ell$) recovers the instantaneous, coherence-limited peak $\omega_{\rm peak} \simeq 1/\ell$ of (85).

C. Single-scale limit: deriving the quadrupole factor of two

The peak tracks whatever scale the source correlations carry; the cleanest limit is a fluid coherent at a single length scale ℓ_0. Here the factor of two relating the GW scale to the fluid scale is not an assumption — it follows from the source being the *quadratic* Reynolds stress $\mathcal{S}_{ij} = w\,v_i v_j$, through Wick's theorem. For a Gaussian velocity field the connected stress two-point function is quadratic in the velocity correlator $R_{ab}(\mathbf{r}) = \langle v_a(\mathbf{r}_1) v_b(\mathbf{r}_2)\rangle$ (the disconnected pairing is the constant mean stress $\langle \mathcal{S}_{ij}\rangle^2$ and does not radiate):

$$\Sigma(\mathbf{r}) = \big\langle \mathcal{S}_{ij}(\mathbf{r}_1)\mathcal{S}_{ij}(\mathbf{r}_2)\big\rangle_{\rm c} = w^2\Big[R_{aa}(\mathbf{r})R_{bb}(\mathbf{r}) + \tfrac{1}{3}R_{ab}(\mathbf{r})R_{ab}(\mathbf{r})\Big] \;\propto\; R(\mathbf{r})^2, \tag{88}$$

the TT-contracted form of Eq. (11); the key feature is that Σ is the *square* of the velocity correlator. A fluid coherent at the single scale ℓ_0 has a velocity correlator that oscillates on that scale: for the isotropic single-scale field the trace is $R_{aa}(r) \propto \sin(r/\ell_0)/(r/\ell_0)$.[10] Squaring *halves the scale*,

$$R(r)^2 \;\propto\; \frac{\sin^2(r/\ell_0)}{(r/\ell_0)^2} = \frac{1-\cos(2r/\ell_0)}{2\,(r/\ell_0)^2}, \tag{89}$$

so the radiating part of $\Sigma(r)$ oscillates on the halved scale $\ell_0/2$ (the constant term is the non-radiating mean). The GW frequency is the inverse of the scale it samples, so the spectrum $d\rho_{\rm GW}/d\ln\omega \propto \omega^3 P(\omega)|\tilde g(\omega)|^2$ collapses to a single line at

$$\boxed{\omega_{\rm GW} = \frac{2}{\ell_0} \qquad \big(\ell_{\rm GW} = \tfrac{1}{2}\,\ell_0\big).} \tag{90}$$

The factor of two is thus *derived*: a quadratic (quadrupole) source $v_i v_j$ radiates on half the scale — twice the frequency — of the field that drives it, Eq. (89) being that halving written as the identity $\sin^2\theta = \frac{1}{2}(1-\cos 2\theta)$. There is no infrared ω^3 rise and no ultraviolet falloff: the ω^3 phase space and the duration filter $|\tilde g(\omega)|^2$ fix only the line's height, so $\omega_{\rm GW}$ is independent of both the firing time t_0 and the duration Δt. This is the configuration-space image of the GW peak at half the source scale found for the monochromatic source in Eq. (74): two velocity features of scale ℓ_0 overlap to produce stress structure on the scale $\ell_0/2$. (A strictly single-scale field in fact fills all scales $\geq \ell_0/2$ — the autoconvolution in Eq. (88) — piling up at the $\ell_0/2$ edge as in Sec. VI; the δ-line keeps only the halved scale.)

VIII. NUMERICAL RESULTS: $f^{\rm GW}_{\rm peak}(\tau_*, M_0)$

We scan the peak GW frequency $f^{\rm GW}_{\rm peak} = \mathrm{argmax}_\omega[\omega\,\Omega_{\rm GW}(\omega)]/2\pi$ over the (τ_*, M_0) plane for a delta-shell decaying source evolved with the BK2016 self-similar closure and time-integrated over the emission window (`Notebooks/f_peak_vs_tau_mac`

Two temporal-correlation kernels are compared through their dimensionless transform $\mathcal{G}(q) = \widehat{R^2}(q)$: the random-phase BK2016 form (Model A) and the coherent eddy (Model B).

Across all four self-similar classes the ratio $f^A_{\rm peak}/f^B_{\rm peak}$ is *independent of* (τ_*, M_0): the self-similar closure makes both the eddy frequency ω_e and the sweeping rate η_0 scale as $(1+T/\tau_*)^{-1}$, so the two kernels inherit the same overall frequency scaling and differ only in their fixed dimensionless peak position. The model difference therefore collapses to a single per-class constant (Fig. 12): $f_A/f_B \simeq 0.37$, 0.36 for the HD (Loitsiansky, Saffman) classes and 0.83, 0.78 for the MHD (nonhelical, helical) classes – Model A always peaks lower. The large HD offset comes from the side-peak of the coherent kernel at the eddy frequency ($q \simeq \pm 2$, the $\cos\sigma$ oscillation of R_B), which wins the $\omega\,\Omega_{\rm GW}$ argmax for the steep-decay HD classes but nearly coincides with the broad Model-A peak for MHD. Because the per-model maps differ only by this constant, we display the ratio rather than the eight individual heatmaps.

IX. DIFFERENCES AMONG THE MODELS

With every kernel of Table I in hand we can now say precisely what distinguishes them, and—more usefully—which of the two axes of Sec. I D each difference belongs to. The summary is Table III; the reasoning follows. Throughout, $\Omega_{\rm GW}(p) \propto p^3 H[S,T](p,p)$ and all numbers are measured by `Notebooks/model_grid_survey.py` at $M = 1$, $R = 10^4$.

a. The infrared belongs to the spatial axis alone. Every kernel with a spatially extended source gives the causal k^3, and every kernel with a monochromatic source gives k^2, *irrespective of the temporal model*: we measure 3.00 for $H[S_{\rm K41}, T_{\rm sw}]$ and 2.97 for $H[S_{\rm K41}, T_{\rm dec}]$, against 2.00 and 1.97 for the two δk kernels and 2.00 for the impulsive one. This is not a coincidence of our kernels but the universal theorem of Sec. IV A: the infrared is fixed by whether the source spectrum has finite width, and a δ-function source violates that hypothesis by exactly one power through the $1/p$ pole of $\mathcal{K}_0(p)/p$. The temporal factor cannot contribute, because as $p \to 0$ it tends to a constant—$\pi\,\mathcal{C}(0) = 6\pi$ for $T_{\rm dec}$ by Eq. (50), unity for $T_{\rm sw}$—and a constant carries no slope. *Corollary:* no choice of UETC can be tuned to reproduce the simulated k^1 of Sec. IV. That tension is not resolvable on the temporal axis, which is precisely why it survives every treatment in Table II.

b. The ultraviolet belongs to the temporal axis alone. Here the roles reverse, and the discriminant is sharp: it is whether the UETC has a *cusp* at zero lag. By Eq. (35) the temporal factor falls as $-2f'(0^+)/\omega^2$, so

- $T_{\rm sw}$ (Gaussian, $f'(0^+) = 0$): no cusp, and the cutoff is super-exponential. Fitting a power law to it is meaningless—the apparent "slope" merely records where the fit window was placed, which is why $H[S_{\delta r}, T_{\rm sw}]$ returns -46.6 and $H[S_{\rm K41}, T_{\rm sw}]$ returns -4.75 on the same window.
- $T_{\rm dec}$ (power law, $f'(0^+) = -4/3$): a genuine heavy tail $\propto \omega^{-2}$, with the coefficient 8/3 of Eq. (50). The measured GW slope is mildly M-dependent (-1.65 at $M = 0.05$ rising to -1.28 at $M = 3$) because the tail sets in only above $q \sim M$.
- $T_{\rm imp}$, $T_{\rm burst}$: white and window-modulated respectively, the latter carrying the $\sin^2$ factor of Eq. (41).

This single dichotomy—cusp or no cusp—is what the caveat of Sec. III B turns on, and it is worth emphasising that it is *not* a matter of degree: the two behaviours differ by fourteen orders of magnitude by $q = 4$.

c. The peak is where the two axes compete. The peak is the one property neither axis owns outright, which is why it has been the most contested quantity in the literature. With $T_{\rm sw}$ the spectrum is cut off at the sweeping frequency and the peak sits at $p \simeq 1.47M$ (measured 1.53 at $M = 1$), *below* the source scale for subsonic turbulence. With $T_{\rm dec}$ the heavy tail survives past that cutoff, and the peak is instead pinned by the spatial structure at $p \simeq 2.3$, essentially independent of M (measured 2.32 for $M \le 1$, drifting to 2.79 by $M = 3$). The two temporal models therefore disagree about the peak by a factor $\sim 1/M$, which for the subsonic simulations is an order of magnitude—this is exactly the Gogoberidze-versus-Roper Pol tension of Sec. III. Note that the δk and δr kernels have no dynamical peak at all: the former is truncated at $p = 2$ by the triangle inequality, and the latter, having $E \propto k^2$, is ultraviolet-dominated and simply rises to the dissipation cutoff. Their tabulated "peaks" are cutoffs, not physics.

d. Positivity is a property of the temporal axis, and not automatic. The temporal factor must be non-negative at every frequency, since it is the power spectrum of a real correlation; a UETC that fails this yields negative gravitational-wave energy. This is not a hypothetical failure mode—it is the defect that led Caprini et al. to discard a Gaussian straining-time model in favour of a top-hat, and that motivates the positive-definite Gibbs kernel of [8]. Both temporal models used here are safe, and for $T_{\rm dec}$ provably so: $(1+s)^{-4/3}$ is completely monotone, hence by Bernstein's theorem a superposition of decaying exponentials, whose cosine transforms $2a/(a^2+\omega^2)$ are individually positive; products of completely monotone functions are completely monotone, so the two-scale factor Eq. (57) inherits the property. We verify $\mathcal{C} > 0$ over $10^{-8} < q < 10^4$.

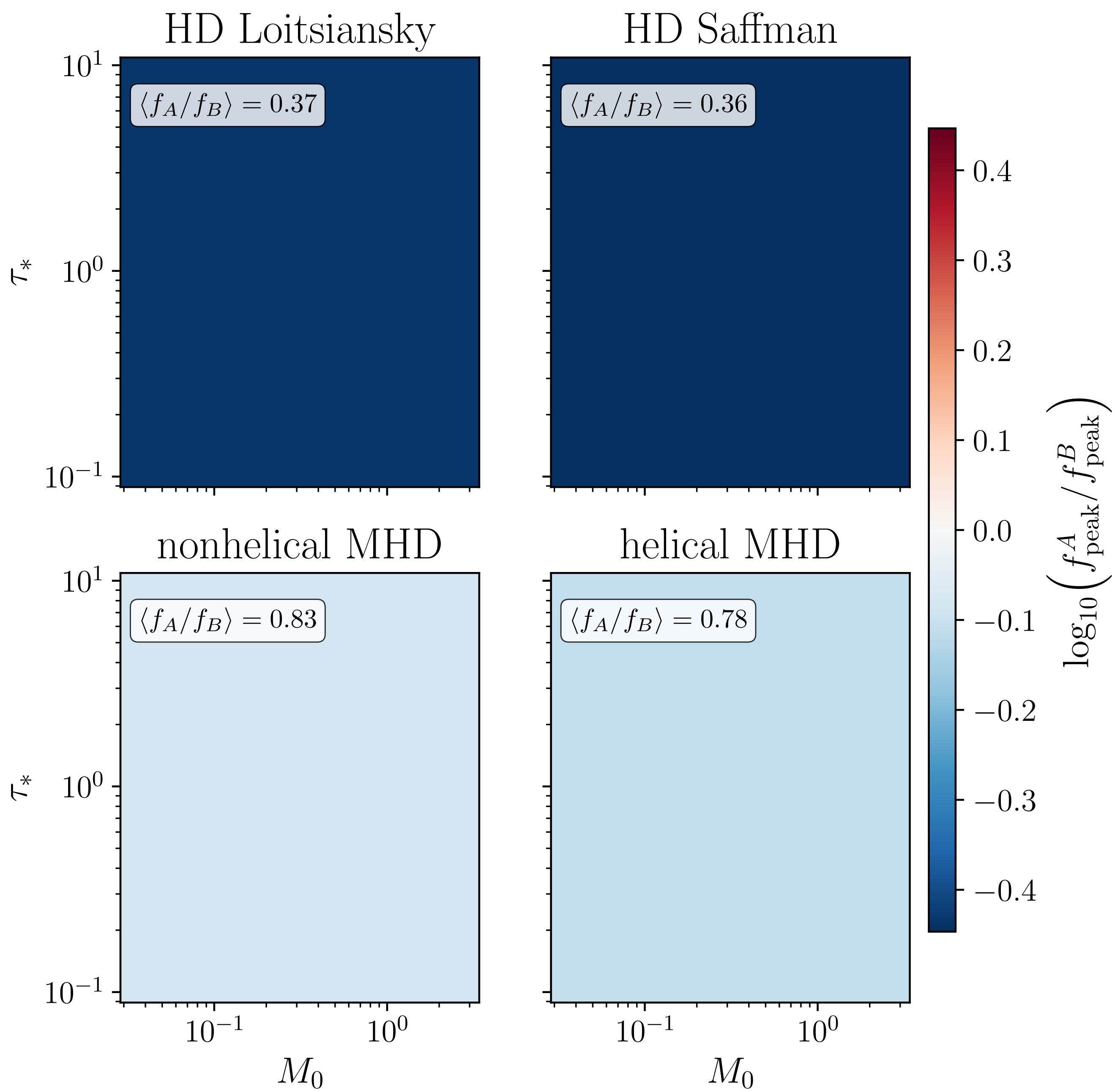


FIG. 12. Model A/B ratio of the peak GW frequency, $\log_{10}(f^A_{\rm peak}/f^B_{\rm peak})$, on the (τ_*, M_0) plane for the four BK2016 self-similar decay classes (HD Loitsiansky $q = 2/7$, $\beta = 4$; HD Saffman $q = 1/3$, $\beta = 3$; nonhelical MHD $q = 1/2$, $\beta = 1$; helical MHD $q = 2/3$, $\beta = 0$). Each panel is flat: the ratio is independent of (τ_*, M_0), fixed only by the kernel shapes, with the per-class value $\langle f_A/f_B\rangle$ annotated. Model A (random-phase) peaks below Model B (coherent eddy) everywhere; the HD classes show the larger offset.

A. The Mach exponent of the peak as a regime diagnostic

The peak's dependence on the Mach number deserves separating out, because it is the one place where the two temporal models make a *qualitatively* different prediction rather than a quantitatively different one. Measuring $d\ln p_{\rm peak}/d\ln M$ at fixed k_0 over $M \in [0.1, 1.6]$ gives

Temporal model	$d \ln p_{\rm peak}/d \ln M$
$T_{\rm sw}$ (Kraichnan sweeping)	$+0.974$
$T_{\rm dec}$ (BK2016 power law)	$+0.048$
$T_{\rm imp}$ (δ-in-time)	0.000

a clean split between unity and zero. The mechanism is transparent: $T_{\rm sw}$ multiplies the spectrum by e^{-q^2/M^2}, a cutoff at $q \sim M$, so the peak simply tracks that cutoff; $T_{\rm dec}$ and $T_{\rm imp}$ impose no such frequency scale, so the peak falls back on the spatial structure. Strikingly, the integrated energy is nearly *blind* to this distinction—we measure $E_{\rm tot} \propto M^{5.8}$ and $M^{5.6}$ for the two—so it is the peak, not the amplitude, that carries the information.

a. What is and is not new here. Both halves of this contrast are already in the literature, and neither is ours. Gogoberidze et al. [2] state that the peak frequency scales linearly with the characteristic velocity, i.e. $f_{\rm peak} \propto M k_0$—our stationary row, of which the coefficient 1.47 of Sec. III is a quantitative refinement. Auclair et al. [8] find instead that the peak is pinned to the integral scale independently of the velocity—our decaying and impulsive rows. What appears not to have been said is that the two sit on a single axis and that the exponent itself is therefore a diagnostic: [8] in particular does not contrast the velocity dependence between decorrelating and coherent sources.

b. What the diagnostic actually diagnoses. It is tempting to read the table as evidence for the power-law UETC specifically, and that reading is wrong. The impulsive kernel contains no M anywhere, so its exponent is identically zero while having no heavy tail, no power law, and indeed no temporal structure at all; and [8] obtain a velocity-independent peak while using Kraichnan *sweeping*, because in a freely decaying flow the integral scale itself evolves. Velocity-independence of the peak is therefore a property of a whole class of sources, not a signature of Eq. (48). The exponent answers "does the temporal factor impose a cutoff at $q \sim M$?"—not "which UETC is it?". Read that way it is a genuine diagnostic; read as support for the decaying model in particular it proves much less than it appears to, and the peak-based argument of Sec. III B should be weighted accordingly.

c. The aeroacoustic precedent. That the Mach scaling is regime-dependent is classical, and the relevant distinction is precisely a temporal one. Lighthill's eighth-power law [11] ($P \propto U^8$, fixed velocity and geometry) and Proudman's [12] ($P \propto \rho\varepsilon M^5$, isotropic turbulence at fixed dissipation rate) differ in what is held fixed—the same ambiguity that flips our own peak scaling between $1.47M$ at fixed k_0 and $1.47/M^2$ at fixed ε. More pointedly, Proudman's M^5 rests on *neglecting retarded-time differences*, replacing the retarded covariance of the stress by the simultaneous one—that is, on treating the source as temporally coherent—and Lilley [13] later rederived the result retaining them. The coherent versus decorrelating axis that separates our two rows is thus the Proudman–Lilley question transposed into a cosmological setting, and our integrated $M^{5.6}$–$M^{5.8}$ sits near Proudman's M^5. The contribution here is not the existence of a regime dependence but its sharpening into a spectral observable: an exponent one reads off the peak rather than off the total power. (The two integrated exponents are approximate; the stationary kernel emits convergence warnings over this sweep.)

d. Analytic structure. Finally, the models differ in what must actually be computed, and the reduction is larger than the usual presentation suggests. The decaying kernels are conventionally written as frequency convolutions of the singular $\hat{g}$, which is both expensive and numerically treacherous; by Eqs. (49) and (57) the single-scale case is closed-form and the two-scale case is a single smooth Gauss–Jacobi quadrature over the closed form. The decaying kernels are therefore no harder than the Gaussian ones.

Kernel	IR slope	Peak p	UV	Positive	Literature counterpart
$H[S_{\rm K41}, T_{\rm sw}]$	3.00	$1.47M$	super-exp	yes	Gogoberidze et al. [2]
$H[S_{\rm K41}, T_{\rm dec}]$	2.97	2.32	ω^{-2}	yes	Roper Pol et al. [6] (peak)
$H[S_{\delta k}, T_{\rm sw}]$	2.00	cutoff	none	yes	—
$H[S_{\delta k}, T_{\rm dec}]$	1.97	cutoff	none	yes	—
$H[S_{\delta r}, T_{\rm sw}]$	3.00	cutoff	super-exp	yes	—
$H[S_{\delta r}, T_{\rm dec}]$	2.97	cutoff	ω^{-2}	yes	—
$H[S_{\delta k}, T_{\rm imp}]$	2.00	cutoff	none	yes	—

TABLE III. Systematic comparison of the kernels of Table I, measured at $M = 1$, $R = 10^4$ on the diagonal $q = p$ by `Notebooks/model_grid_survey.py`. The infrared slope tracks the *spatial* axis only (3 for extended sources, 2 for monochromatic ones, per Sec. IV A); the ultraviolet tracks the *temporal* axis only (a power-law tail iff the UETC has a cusp at zero lag, Eq. (35)); only the peak depends on both. "cutoff" denotes models with no dynamical peak—the δk kernels are truncated at $p = 2$ by the triangle inequality and the δr kernels are ultraviolet-dominated. "super-exp" denotes a Gaussian cutoff, for which no power-law slope exists.

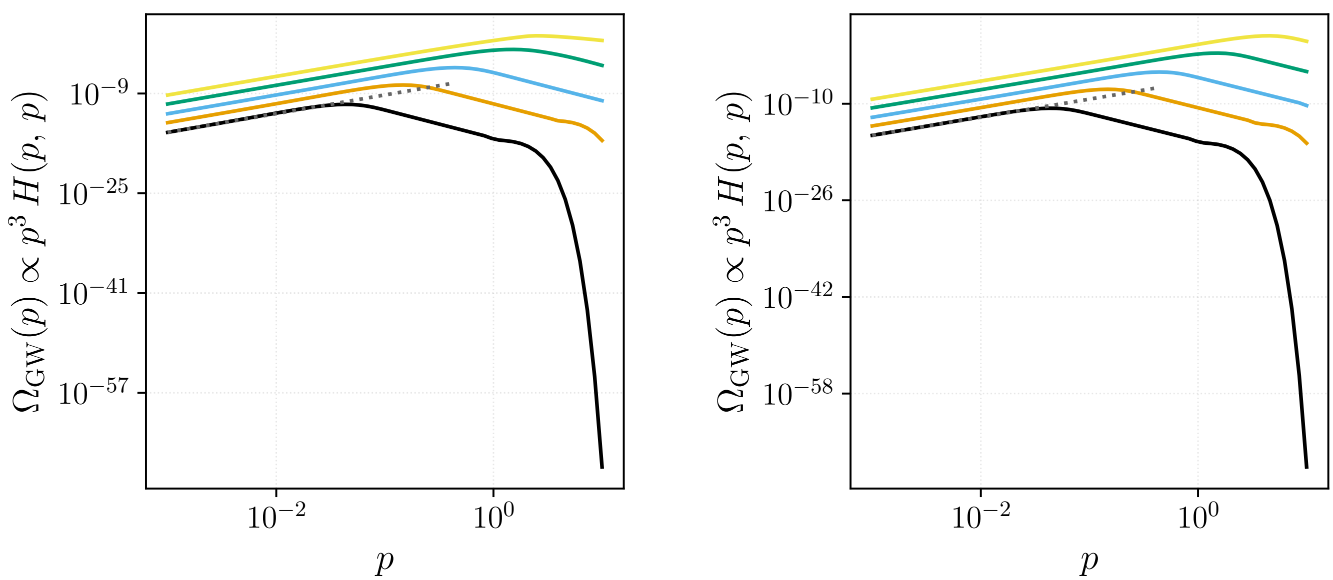
X. SPECTRUM SURVEY FOR ALL DERIVED MODELS

In this section we plot, on a common convention, the dimensionless GW energy-density spectrum

$$\Omega_{\rm GW}(p) \;\propto\; p^3\, H(p,p), \qquad p = k_{\rm GW}/k_0, \tag{91}$$

along the sound-cone diagonal $q = p$ for every model derived in this paper, in the uniform notation $H[S,T]$ of Sec. I D. The p^2 factor relative to the characteristic strain $h_c^2(p) = p\,H(p,p)$ is the phase-space measure d^3k; together with causality and a source of finite spectral width it predicts $\Omega_{\rm GW} \sim p^3$ in the IR, followed by a peak and a decay in the UV. A dotted grey line shows the p^3 reference.

These plots are the graphical form of Table III, and should be read with its two organising facts in mind: the infrared slope is fixed by the *spatial* model alone and the ultraviolet by the *temporal* model alone. In particular the monochromatic panels sit at p^2 rather than p^3 not because of anything in their temporal treatment but because a zero-width source carries a $1/p$ pole, $\mathcal{K}_0(p)/p$, which is exactly the δ-function exception to the universal infrared theorem reproduced in Sec. IV A. The build log of `Notebooks/all_spectra.py` flags cases whose IR slope departs from 3 by more than ± 0.5; the monochromatic models are expected to be flagged, and are not anomalies.



(a) full H_{pq} (b) $p \to 0$ limit

FIG. 13. Kraichnan decorrelation + Kolmogorov spectrum. Five curves per panel correspond to $M = 0.03$, 0.1, 0.3, 1.0, 3.0 at $R = 10^4$. Panel (a) is the full numerical H_{pq} (Sec. II); panel (b) is the aeroacoustic $p \to 0$ limit (Kahniashvili et al. 2008). Both give the universal IR slope $\Omega_{\rm GW} \propto p^3$.

Appendix A: Table of symbols

[1] A. Kosowsky, A. Mack, and T. Kahniashvili, Phys. Rev. D **66**, 024030 (2002), arXiv:astro-ph/0111483 [astro-ph].
[2] G. Gogoberidze, T. Kahniashvili, and A. Kosowsky, Phys. Rev. D **76**, 083002 (2007), arXiv:0705.1733 [astro-ph].
[3] T. Kahniashvili, L. Campanelli, G. Gogoberidze, Y. Maravin, and B. Ratra, Phys. Rev. D **78**, 123006 (2008), [Erratum: Phys.Rev.D 79, 109901 (2009)], arXiv:0809.1899 [astro-ph].
[4] C. Caprini and D. G. Figueroa, Class. Quant. Grav. **35**, 163001 (2018), arXiv:1801.04268 [astro-ph.CO].
[5] M. Maggiore, *Gravitational Waves. Vol. 1: Theory and Experiments* (Oxford University Press, 2007).
[6] A. Roper Pol, S. Mandal, A. Brandenburg, T. Kahniashvili, and A. Kosowsky, Phys. Rev. D **102**, 083512 (2020), arXiv:1903.08585 [astro-ph.CO].

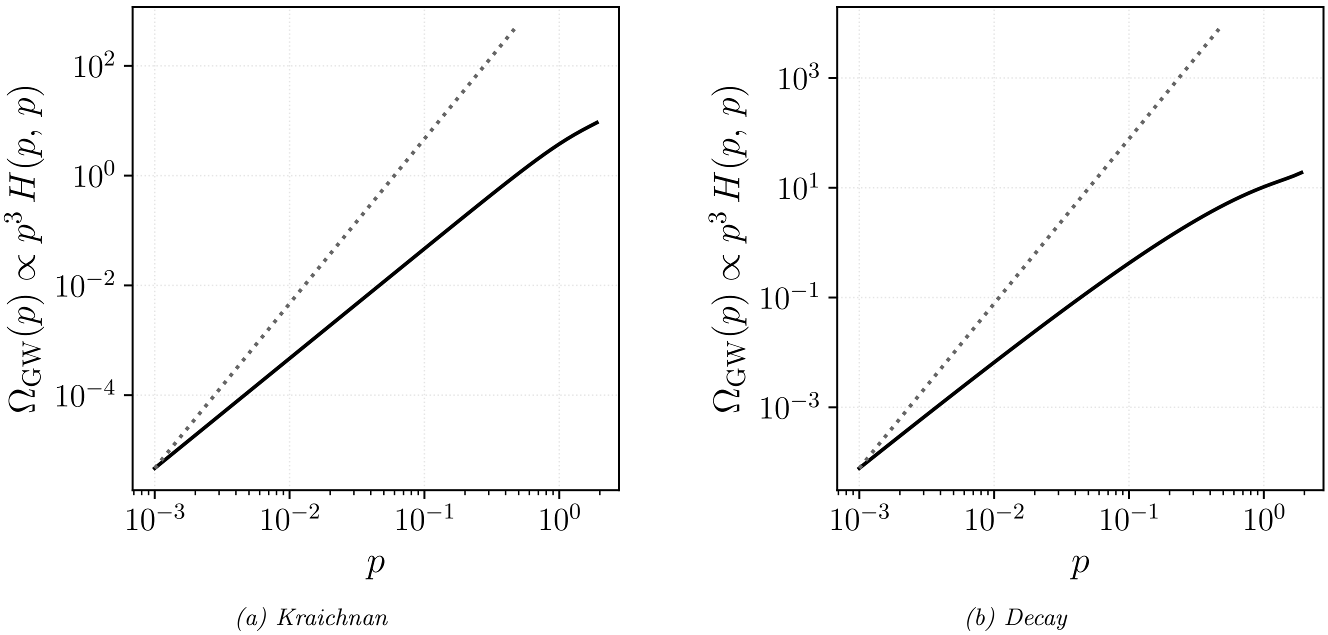


FIG. 14. Monochromatic spectrum $E(k) = E_0\delta(k-k_0)$ (no free parameters at the dimensionless level). Panel (a): Kraichnan decorrelation, closed form $\Omega_{\rm GW}(p) \propto p^3\,K_0(p)/p\,\Theta(2-p)\,e^{-p^2/(2\pi)}$ with IR slope **2** (flagged: $1/p$ pole from the monochromatic source removes one power of p from the universal scaling). Panel (b): decay temporal factor, Eq. (49), IR slope **2** — the *same* slope as panel (a), as it must be, since the temporal factor tends to the constant $\pi\,\mathcal{C}(0)$ as $p \to 0$ and so cannot alter the infrared scaling.

[7] T. Kahniashvili, A. Brandenburg, G. Gogoberidze, S. Mandal, and A. Roper Pol, Physical Review Research **3** (2021), 10.1103/physrevresearch.3.013193.

[8] P. Auclair, C. Caprini, D. Cutting, M. Hindmarsh, K. Rummukainen, D. A. Steer, and D. J. Weir, JCAP **09**, 029 (2022), arXiv:2205.02588 [gr-qc].

[9] R.-G. Cai, S. Pi, and M. Sasaki, Phys. Rev. D **102**, 083528 (2020), arXiv:1909.13728 [astro-ph.CO].

[10] This is the standard spectral-tensor result for homogeneous isotropic turbulence [14–17]. Incompressibility fixes the velocity correlator's tensor structure, leaving its radial part set by a single scalar; the trace transforms as $R_{aa}(r) = 2\int_0^\infty E(k)\,\frac{\sin kr}{kr}\,dk$ with E the energy spectrum, so a fluid carrying a single scale (E concentrated at one $1/\ell_0$) gives $R_{aa}(r) \propto \sin(r/\ell_0)/(r/\ell_0)$ and $R_{aa}(0) = \langle v^2\rangle$. Physically $\sin(r/\ell_0)/(r/\ell_0)$ is the directional average of a plane wave over the sphere of fixed scale — the correlation of a random field built from equal-size eddies pointing every way: it first vanishes at $r = \pi\ell_0$ (points half a wavelength apart are anti-correlated) and decays as ℓ_0/r as the phases wash out. The same angular reduction underlies the GW-from-turbulence kernels [1, 2].

[11] M. J. Lighthill, Proc. R. Soc. Lond. A **211**, 564 (1952).

[12] I. Proudman, Proc. R. Soc. Lond. A **214**, 119 (1952).

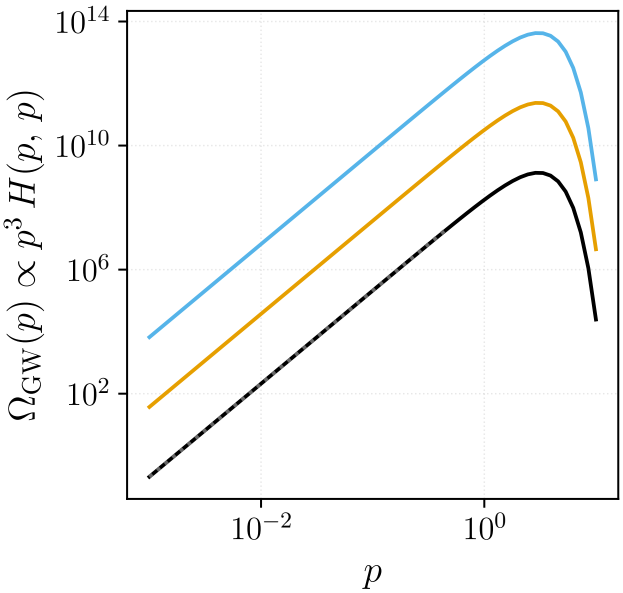
[13] G. M. Lilley, Theor. Comput. Fluid Dyn. **6**, 281 (1994).

[14] G. K. Batchelor, *The Theory of Homogeneous Turbulence* (Cambridge University Press, Cambridge, 1953).

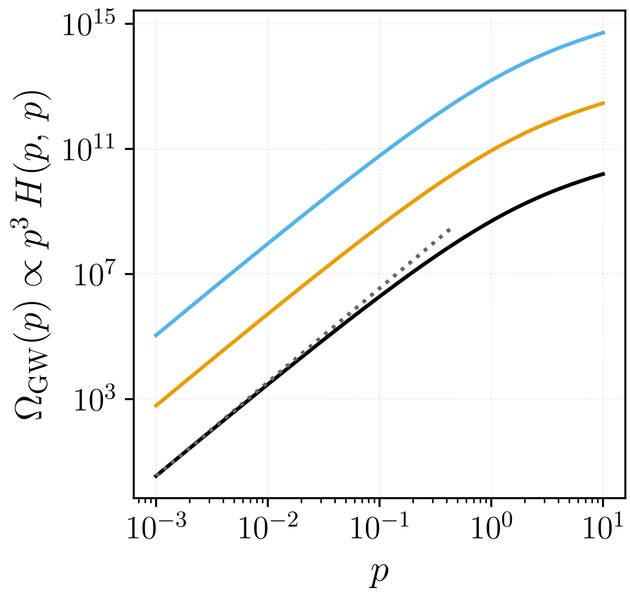
[15] S. B. Pope, *Turbulent Flows* (Cambridge University Press, Cambridge, 2000).

[16] A. S. Monin and A. M. Yaglom, *Statistical Fluid Mechanics: Mechanics of Turbulence, Volume 2* (MIT Press, Cambridge, MA, 1975).

[17] R. H. Kraichnan, Journal of Fluid Mechanics **5**, 497 (1959).



(a) Kraichnan



(b) Decay

FIG. 15. White-noise real-space correlations $R_{ij} \propto \delta^{(3)}(\mathbf{r})$. Three curves per panel correspond to $R = 10^3$, 10^4, 10^5. Panel (a): Kraichnan decorrelation, matches the universal IR slope **3**. Panel (b): decay temporal model, IR slope **3** as well, for all three R — the decaying and Kraichnan models share the universal causal infrared, the temporal factor affecting only the amplitude and the ultraviolet tail.